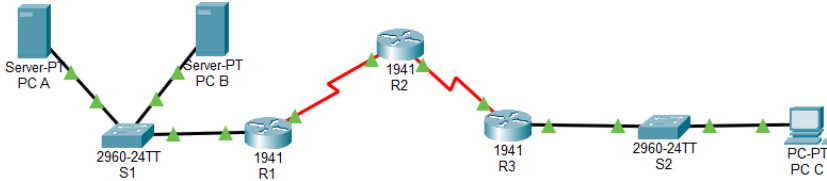
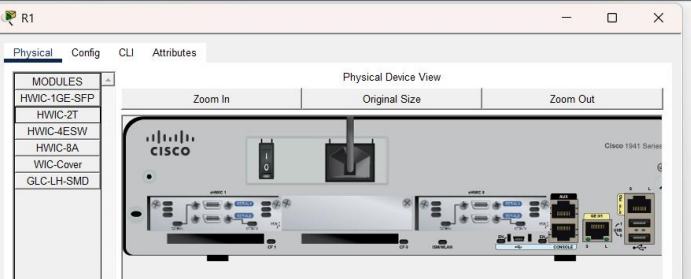


V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

Practical 1	Packet Tracer - Configure Cisco Routers for Syslog, NTP, and SSH Operations PC A to Switch 1 Fastethernet 0/1 PC B to Switch 1 Fastethernet 0/2 Switch 2 to PC C fastethernet 0/1 R1 to switch g0/1 fasethernet 0/3 Choose wire serial dce R1 -> R2 s0/0 Use wire serial dte R2->R3 serail 0/0/1 Use copper wire R3 -> s1 g0/1
STEPS	Set up by taking necessary server, router and PCs USE COPPER WIRE 
	 For all three routers, Go to physical ->Power off-> Select HWIC-2T and place 2 chips.
	Setting up addresses

V. G. VAZE KELKAR COLLEGE, MULUND (E)

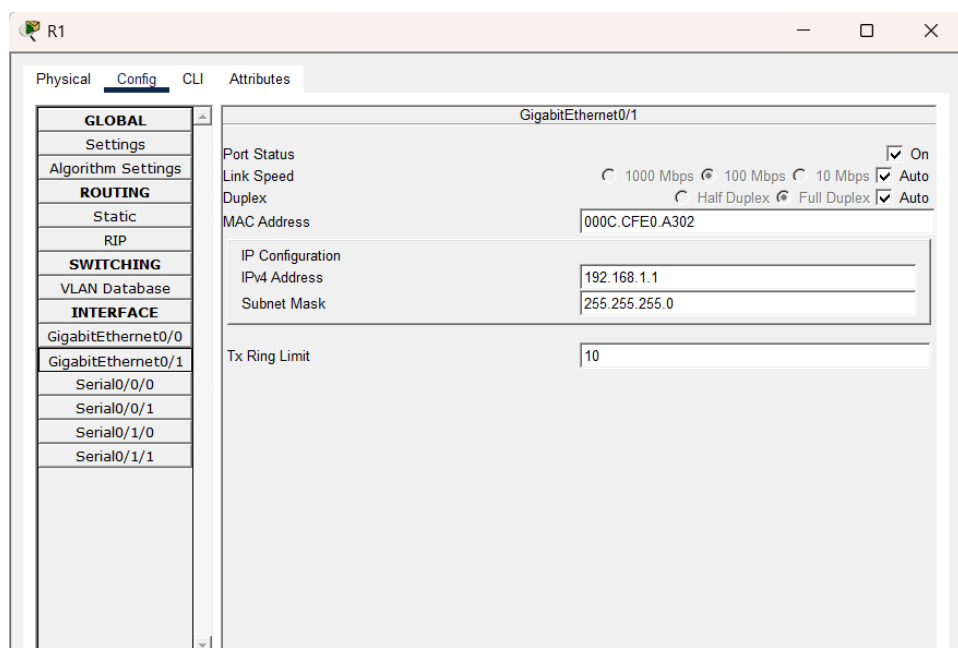
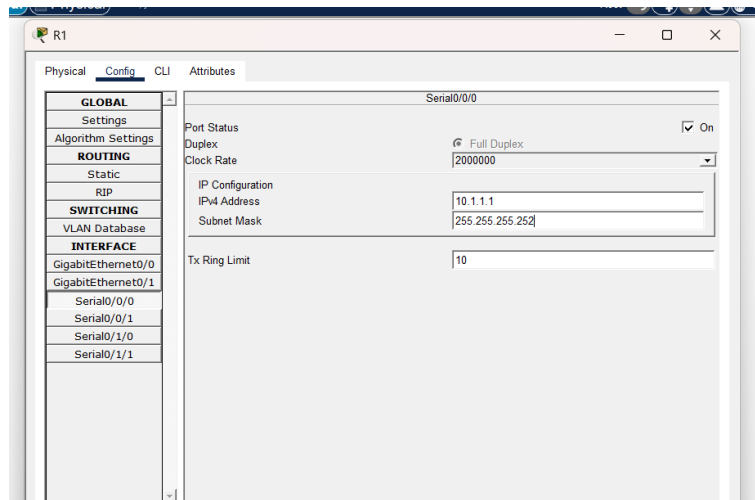
Roll No: A004

Name: Dhruv N Chandalia

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

for R1



Equivalent IOS Commands

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

Setting up IP addresses for R2

The screenshot shows the configuration window for Router R2, specifically the 'Config' tab for the 'Serial0/0/1' interface. The left sidebar shows the configuration tree with 'Serial0/0/1' selected under the 'INTERFACE' section. The main area displays the following settings:

- Port Status: ☒ On
- Duplex: ☒ Full Duplex
- Clock Rate: 2000000
- IP Configuration:
 - IPv4 Address: 10.2.2.2
 - Subnet Mask: 255.255.255.252
- Tx Ring Limit: 10

Below the configuration area, the 'Equivalent IOS Commands' are listed:

```
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#no shutdown
Router(config-if)#ip address 10.1.1.2 255.0.0.0
Router(config-if)#ip address 10.1.1.2 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#no shutdown
Router(config-if)#ip address 10.2.2.2 255.255.255.252
Router(config-if)#
```

The screenshot shows the configuration window for Router R2, specifically the 'Config' tab for the 'Serial0/0/0' interface. The left sidebar shows the configuration tree with 'Serial0/0/0' selected under the 'INTERFACE' section. The main area displays the following settings:

- Port Status: ☒ On
- Duplex: ☒ Full Duplex
- Clock Rate: 2000000
- IP Configuration:
 - IPv4 Address: 10.1.1.2
 - Subnet Mask: 255.255.255.252
- Tx Ring Limit: 10

Below the configuration area, the 'Equivalent IOS Commands' are listed:

```
Router(config)#no shutdown
Router(config-if)#ip address 10.1.1.2 255.0.0.0
Router(config-if)#ip address 10.1.1.2 255.255.255.252
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#no shutdown
```

Setting up addresses for R3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R3

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

GigabitEthernet0/1

Port Status ☒ On

Link Speed 1000 Mbps 100 Mbps 10 Mbps ☒ Auto

Duplex Half Duplex Full Duplex ☒ Auto

MAC Address 0001.96D9.DB02

IP Configuration

IPv4 Address 192.168.3.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

R3

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

Serial0/0/1

Port Status ☒ On

Duplex Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.2.2.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

For PC A

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

PC A

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.5

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::203:E4FF:FEBE:BD0

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

For PC B

PC B

Physical Config Services **Desktop** Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.6

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:43FF:FECD:74D0

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

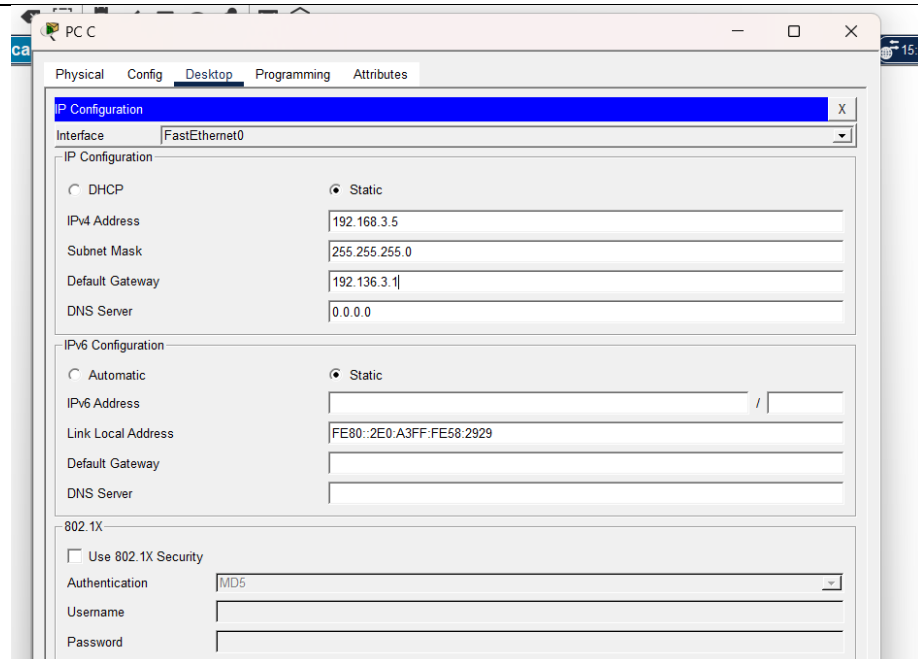
Password

For PC C

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



Performing ospf---
Router 1 config

```
r1>enable
r1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#router ospf 1
r1(config-router)#network 10.1.1.0 0.0.0.3 area 0
r1(config-router)#network 192.168.1.0 0.0.0.255 area 0
r1(config-router)#end
r1#
%SYS-5-CONFIG_I: Configured from console by console

r1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
10.2.2.2        0    FULL/ -         00:00:35    10.1.1.2       Serial0/0/0

r1#show ip route ospf
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
O        10.2.2.0 [110/128] via 10.1.1.2, 00:03:21, Serial0/0/0
O    192.168.3.0 [110/129] via 10.1.1.2, 00:03:21, Serial0/0/0

r1#
```

Copy

Paste

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

%SYS-5-CONFIG_I: Configured from console by console

```
r1#enable
r1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#interface GigabitEthernet0/1
r1(config-if)#no shutdown
r1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
r1(config-if)#show interfaces status
^
% Invalid input detected at '^' marker.
r1(config-if)#exit
r1(config)#exit
r1#
%SYS-5-CONFIG_I: Configured from console by console
r1#show interfaces status
^
% Invalid input detected at '^' marker.
r1#enable
r1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#show interfaces status
^
```

Copy

Paste

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
r1(config-if)#show interfaces status
^
% Invalid input detected at '^' marker.
r1(config-if)#exit
r1(config)#exit
r1#
%SYS-5-CONFIG_I: Configured from console by console
r1#show interfaces status
^
% Invalid input detected at '^' marker.
r1#enable
r1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r1(config)#show interfaces status
^
% Invalid input detected at '^' marker.
r1(config)#show interface brief
^
% Invalid input detected at '^' marker.
r1(config)#exit
r1#
%SYS-5-CONFIG_I: Configured from console by console
r1#show interface brief
^
% Invalid input detected at '^' marker.
r1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 192.168.1.1 YES manual up up
Serial0/0/0 10.1.1.1 YES manual up up
Serial0/0/1 unassigned YES unset administratively down down
Serial0/1/0 unassigned YES unset administratively down down
Serial0/1/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
r1#
```

Router 2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
%Error parsing filename (Bad file number)
r2(config-if)#conf t
%Invalid hex value
r2(config)#router ospf 1
r2(config-router)#network 10.1.1.0 0.0.0.3 area 0
r2(config-router)#network 10.2.2.0 0.0.0.
00:12:38: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL,
Loading Done

      ^
% Invalid input detected at '^' marker.

r2(config-router)#network 10.2.2.0 0.0.0.3 area 0
r2(config-router)#exit
r2(config)#exit
r2#
%SYS-5-CONFIG_I: Configured from console by console

r2#show ip route ospf
O    192.168.1.0 [110/65] via 10.1.1.1, 00:00:24, Serial0/0/0
r2#
```

```
r2>enable
r2#show ip route ospf
O    192.168.3.0 [110/65] via 10.2.2.1, 00:06:10, Serial0/0/1

r2#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	administratively down	down
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.1.1.2	YES	manual	up	up
Serial0/0/1	10.2.2.2	YES	manual	up	up
Serial0/1/0	unassigned	YES	unset	administratively down	down
Serial0/1/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
r2#
```

Router 3

```
% Invalid input detected at '^' marker.

r3>enable
r3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
r3(config)#interface g0/1
r3(config-if)#ip address 192.168.3.1 255.255.255.0
r3(config-if)#no shutdown
r3(config-if)#exit
r3(config)#interface s0/0/1
r3(config-if)#ip address 10.2.2.1 255.255.255.252
r3(config-if)#no shutdown

r3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

r3(config-if)#exit
r3(config)#router ospf
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
1
r3(config-router)#router ospf 1
r3(config-router)#network 192.168.3.0 0.0.0.255 area 0
r3(config-router)#network 10.2.2.0 0.0.0.3 area 0
r3(config-router)#exit
r3(config)#
00:34:31: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from LOADING to FULL,
Loading Done
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>r3(config)#exit r3# %SYS-5-CONFIG_I: Configured from console by console r3#show ip interface brief Interface IP-Address OK? Method Status Protocol GigabitEthernet0/0 unassigned YES unset administratively down down GigabitEthernet0/1 192.168.3.1 YES manual up up Serial0/0/0 unassigned YES unset administratively down down Serial0/0/1 10.2.2.1 YES manual up up Serial0/1/0 unassigned YES unset administratively down down Serial0/1/1 unassigned YES unset administratively down down Vlan1 unassigned YES unset administratively down down r3#show ip ospf route ^ % Invalid input detected at '^' marker. r3#show ip route ospf 10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks O 10.1.1.0 [110/128] via 10.2.2.2, 00:00:57, Serial0/0/1 O 192.168.1.0 [110/129] via 10.2.2.2, 00:00:47, Serial0/0/1 r3#</pre>
	Verify that messages are successfully delivered.
Step 2 & 3	Configure OSPF MD5 authentication for all routers in area 0. Configure OSPF MD5 authentication for all routers in area 0. Configure an MD5 key on the serial interfaces on R1, R2 and R3. Use the password MD5pa55 for key 1. Router 1 <pre>Enter configuration commands, one per line. End with CNTL/Z. r1(config)#router ospf 1 r1(config-router)#area 0 authentication message-digest r1(config-router)# 00:58:59: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired 00:58:59: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached 00:59:19: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL, Loading Done r1(config-router)#interface s0/0/0 r1(config-if)#ip ospf message-digest-key 1 md5 MD5pa55 r1(config-if)#</pre> Router 2 <pre>r2>enable r2#conf t Enter configuration commands, one per line. End with CNTL/Z. r2(config)#router ospf 1 r2(config-router)#area 0 authentication message-sigest 01:17:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired 01:17:20: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached ^ % Invalid input detected at '^' marker. r2(config-router)#area 0 authentication message-digest r2(config-router)# r2(config-router)#interface s0/0/1 r2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55 r2(config-if)# 01:20:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from LOADING to FULL, Loading Done</pre> Router 3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

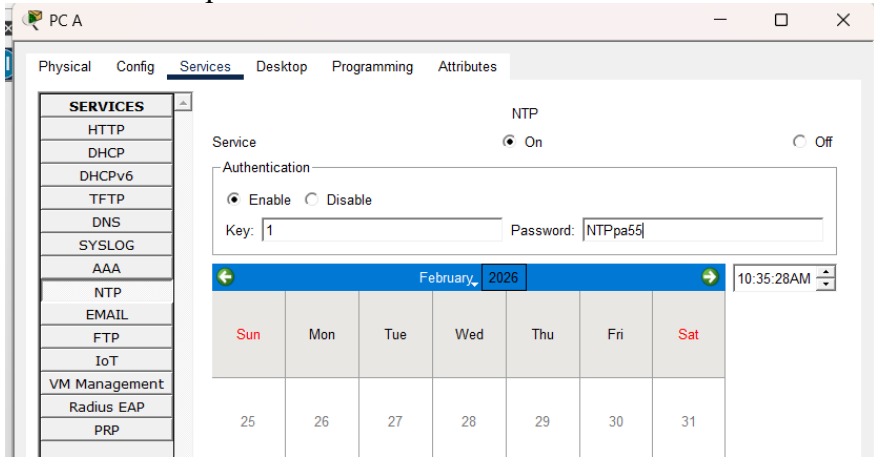
Name: Dhruv N Chandalia

	<pre>r3>enable r3#conf t Enter configuration commands, one per line. End with CNTL/Z. r3(config)#router ospf 1 r3(config-router)#area 0 authentication message-digest ^ % Invalid input detected at '^' marker. r3(config-router)# 01:18:02: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from FULL to DOWN, Neighbor Down: Dead timer expired 01:18:02: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from FULL to DOWN, Neighbor Down: Interface down or detached r3(config-router)#area 0 authentication message-digest r3(config-router)# 01:18:22: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from LOADING to FULL, Loading Done r3(config-router)#interface s0/0/1 r3(config-if)#ip ospf message-digest-key 1 md5 MD5pa55 r3(config-if)#</pre>
Step 4	Verify configurations Router 1 <pre>HELLO due in 00:00:00 r1#show ip ospf interface s0/0/0 Serial0/0/0 is up, line protocol is up Internet address is 10.1.1.1/30, Area 0 Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT, Cost: 64 Transmit Delay is 1 sec, State POINT-TO-POINT, Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5 Hello due in 00:00:04 Index 2/2, flood queue length 0 Next 0x0(0)/0x0(0) Last flood scan length is 1, maximum is 1 Last flood scan time is 0 msec, maximum is 0 msec Suppress hello for 0 neighbor(s) Message digest authentication enabled Youngest key id is 1 r1#</pre> Router 2 <pre>r2>enable r2#show ip ospf interface Serial0/0/1 is up, line protocol is up Internet address is 10.2.2.2/30, Area 0 Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64 Transmit Delay is 1 sec, State POINT-TO-POINT, Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5 Hello due in 00:00:03 Index 1/1, flood queue length 0 Next 0x0(0)/0x0(0) Last flood scan length is 1, maximum is 1 Last flood scan time is 0 msec, maximum is 0 msec Neighbor Count is 1, Adjacent neighbor count is 1 Adjacent with neighbor 192.168.3.1 Suppress hello for 0 neighbor(s) Message digest authentication enabled Youngest key id is 1 Serial0/0/0 is up, line protocol is up Internet address is 10.1.1.2/30, Area 0 Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64 Transmit Delay is 1 sec, State POINT-TO-POINT, Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5 Hello due in 00:00:03 --More--</pre> Router 3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>r3#show run section router ospf router ospf 1 log-adjacency-changes area 0 authentication message-digest network 192.168.3.0 0.0.0.255 area 0 network 10.2.2.0 0.0.0.3 area 0 r3#show ip ospf interface GigabitEthernet0/1 is up, line protocol is up Internet address is 192.168.3.1/24, Area 0 Process ID 1, Router ID 192.168.3.1, Network Type BROADCAST, Cost: 1 Transmit Delay is 1 sec, State DR, Priority 1 Designated Router (ID) 192.168.3.1, Interface address 192.168.3.1 No backup designated router on this network Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5 Hello due in 00:00:08 Index 1/1, flood queue length 0 Next 0x0(0)/0x0(0) Last flood scan length is 1, maximum is 1 Last flood scan time is 0 msec, maximum is 0 msec Neighbor Count is 0, Adjacent neighbor count is 0 Suppress hello for 0 neighbor(s) Message digest authentication enabled No key configured, using default key id 0 Serial0/0/1 is up, line protocol is up Internet address is 10.2.2.1/30, Area 0 Process ID 1, Router ID 192.168.3.1, Network Type POINT-TO-POINT, Cost: 64 Transmit Delay is 1 sec, State POINT-TO-POINT, Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5 --More--</pre>
Part 2	Configure NTP
Step 1	On PC A -> Services -> NTP -> authentication Select enable Key :1 Password: NTPpa55 
Step 2	Configure R1, R2 and R3 <pre>Youngest key id is 1 r1#enable r1#conf t Enter configuration commands, one per line. End with CNTL/Z. r1(config)#ntp server 192.168.1.5 r1(config)#ntp update-calendar r1(config)#ntp authenticate r1(config)#ntp trusted-key 1 r1(config)#ntp authentication-key 1 md5 NTPa55 r1(config)#service timestamps log datetime msec r1(config)#</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>r2#conf t Enter configuration commands, one per line. End with CNTL/Z. r2(config)#ntp server 192.168.1.5 r2(config)#ntp update-calendar r2(config)#ntp authenticate r2(config)#ntp trusted-key 1 r2(config)#ntp authentication-key 1 md5 NTPpa55 r2(config)#service timestamps log datetime msec r2(config)#</pre> <pre>r3#conf t Enter configuration commands, one per line. End with CNTL/Z. r3(config)#ntp server 192.168.1.5 r3(config)#ntp update-calendar r3(config)#ntp authenticate r3(config)#ntp trusted-key 1 r3(config)#ntp authentication-key 1 md5 NTPpa55 r3(config)#service timestamps log datetime msec r3(config)#</pre>
Part 3	Configure Routers to log messages to the syslog server
	<pre>R1(config)# logging host 192.168.1.6 R2(config)# logging host 192.168.1.6 R3(config)# logging host 192.168.1.6</pre> Router 1 <pre>r1(config)#ntp authentication-key 1 md5 NTPa55 r1(config)#service timestamps log datetime msec r1(config)#logging host 192.168.1.6 r1(config)#exit r1# *Feb 07, 10:52:30.5252: SYS-5-CONFIG I: Configured from console by console *Feb 07, 10:52:30.5252: *SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.6 port 514 started - CLI initiated r1#show logging Syslog logging: enabled (0 messages dropped, 0 messages rate-limited, 0 flushes, 0 overruns, xml disabled, filtering disabled) No Active Message Discriminator. No Inactive Message Discriminator. Console logging: level debugging, 32 messages logged, xml disabled, filtering disabled Monitor logging: level debugging, 32 messages logged, xml disabled, filtering disabled Buffer logging: level debugging, 0 messages logged, xml disabled, filtering disabled Logging Exception size (4096 bytes) Count and timestamp logging messages: disabled Persistent logging: disabled No active filter modules. --More--</pre> Copy Paste Router 2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
r2(config)#ntp authentication-key 1 md5 NTPpa55
r2(config)#service timestamps log datetime msec
r2(config)#logging host 192.168.1.6
r2(config)#show logging
^
% Invalid input detected at '^' marker.

r2(config)#exit
r2#
*Mar 01, 02:21:12.2121: SYS-5-CONFIG I: Configured from console by console
*Mar 01, 02:21:12.2121: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.6 port 514
started - CLI initiated
r2#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 44 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 44 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

--More--
```

Router 3

```
r3(config)#service timestamps log datetime msec
r3(config)#logging host 192.168.1.6
r3(config)#exit
r3#
*Mar 01, 02:28:11.2828: SYS-5-CONFIG I: Configured from console by console
*Mar 01, 02:28:11.2828: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.6 port 514
started - CLI initiated
r3#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 29 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 29 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

--More--
```

Copy

Paste

Part 4

Configure R3 to support SSH connections
(tried deleting sshadmin user and then created disha)

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
r3#enable
r3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r3(config)#no ip ssh
% Incomplete command.
r3(config)#no username SSHadmin
r3(config)#line vty 0 4
r3(config-line)#no login
r3(config-line)#transport input none
r3(config-line)#exit
r3(config)#crypto key zeroize rsa
% All RSA keys will be removed.
% All router certs issued using these keys will also be removed.
Do you really want to remove these keys? [yes/no]: yes
r3(config)#end
*Mar 1 2:42:56.699: %SSH-5-DISABLED: SSH 1.5 has been disabled
r3#
*Mar 01, 02:42:58.4242: SYS-5-CONFIG_I: Configured from console by console
r3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
r3(config)#hostname R3
R3(config)#ip domain-name ccnasecurity.com
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#ip ssh version 2
*Mar 1 2:43:40.934: %SSH-5-ENABLED: SSH 2 has been enabled
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#username disha privilege 15 secret abc55
R3(config)#line vty 0 4
```

```
R3(config)#ip domain-name ccnasecurity.com
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#ip ssh version 2
*Mar 1 2:43:40.934: %SSH-5-ENABLED: SSH 2 has been enabled
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#username disha privilege 15 secret abc55
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
R3(config)#end
R3#
*Mar 01, 02:44:53.4444: SYS-5-CONFIG_I: Configured from console by console
R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R3#show run | section line vty
line vty 0 4
  login local
  transport input ssh
R3#show run | include username
username disha privilege 15 secret 5 $1$mERr$HjTpABk6CJpH2bAziReRF0
R3#
```

In PC C command prompt ->

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre> [Connection to 192.168.3.1 closed by foreign host] C:\>ssh -l disha 192.168.3.1 Password: R3# </pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

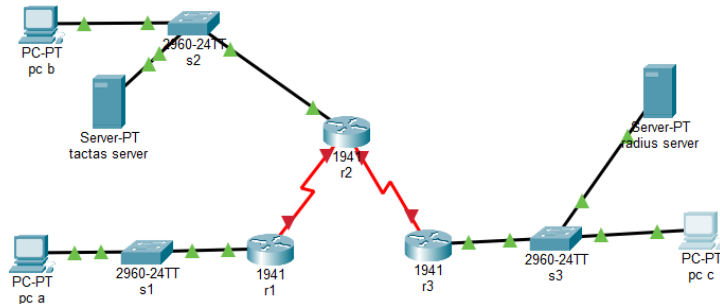
Roll No: A004

Name: Dhruv N Chandalia

PRACTICAL 2

Packet Tracer - Configure AAA Authentication on Cisco Routers

Topology



Addressing Table

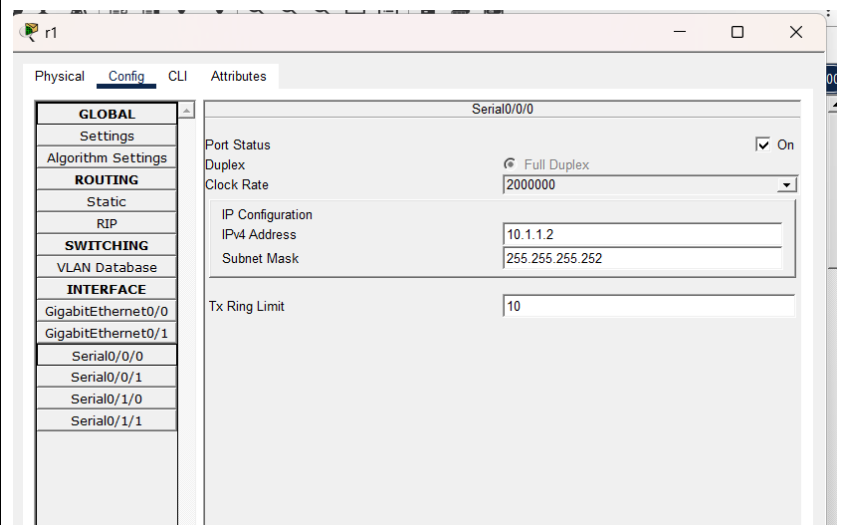
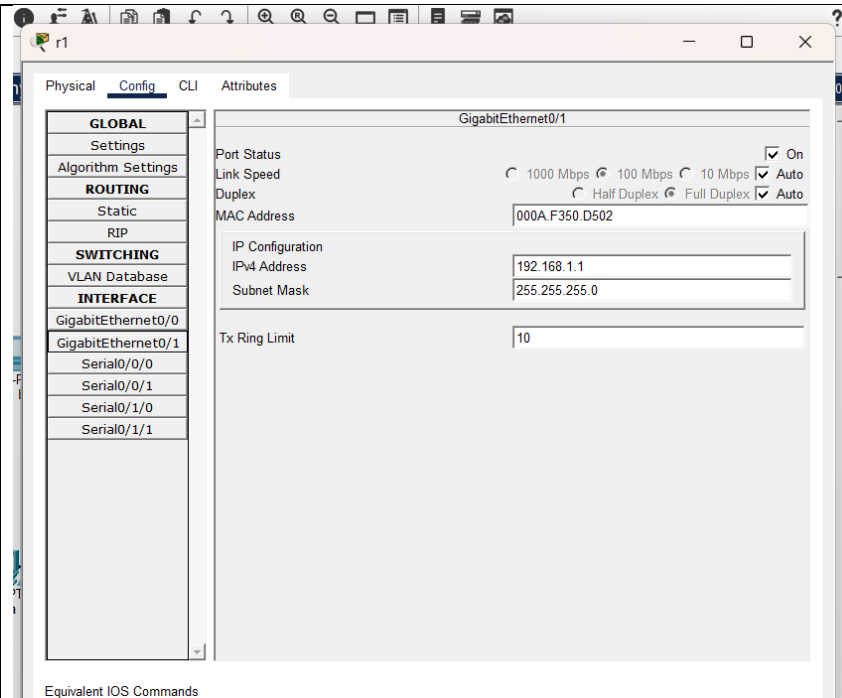
Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

For Router 1

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

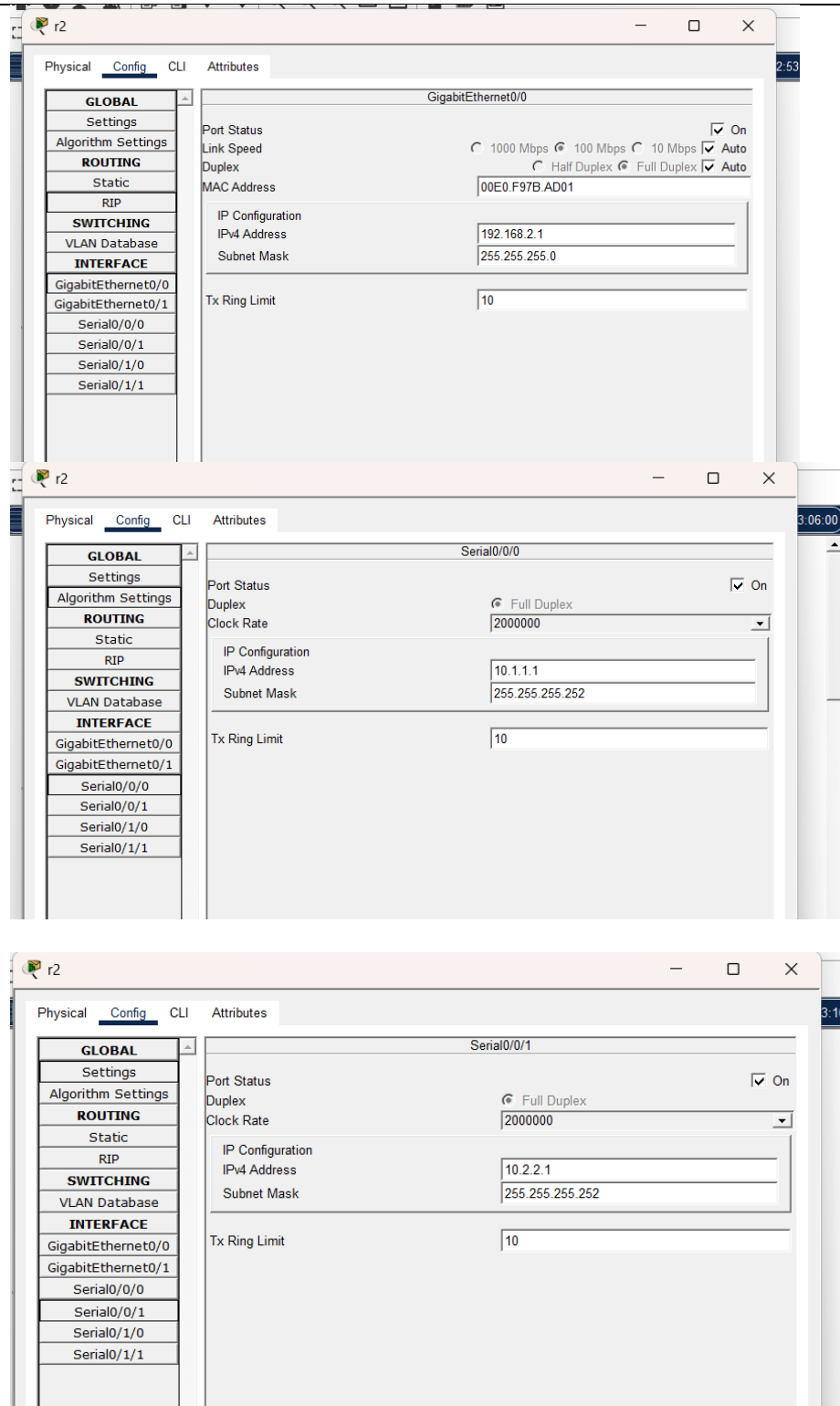


For Router 2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

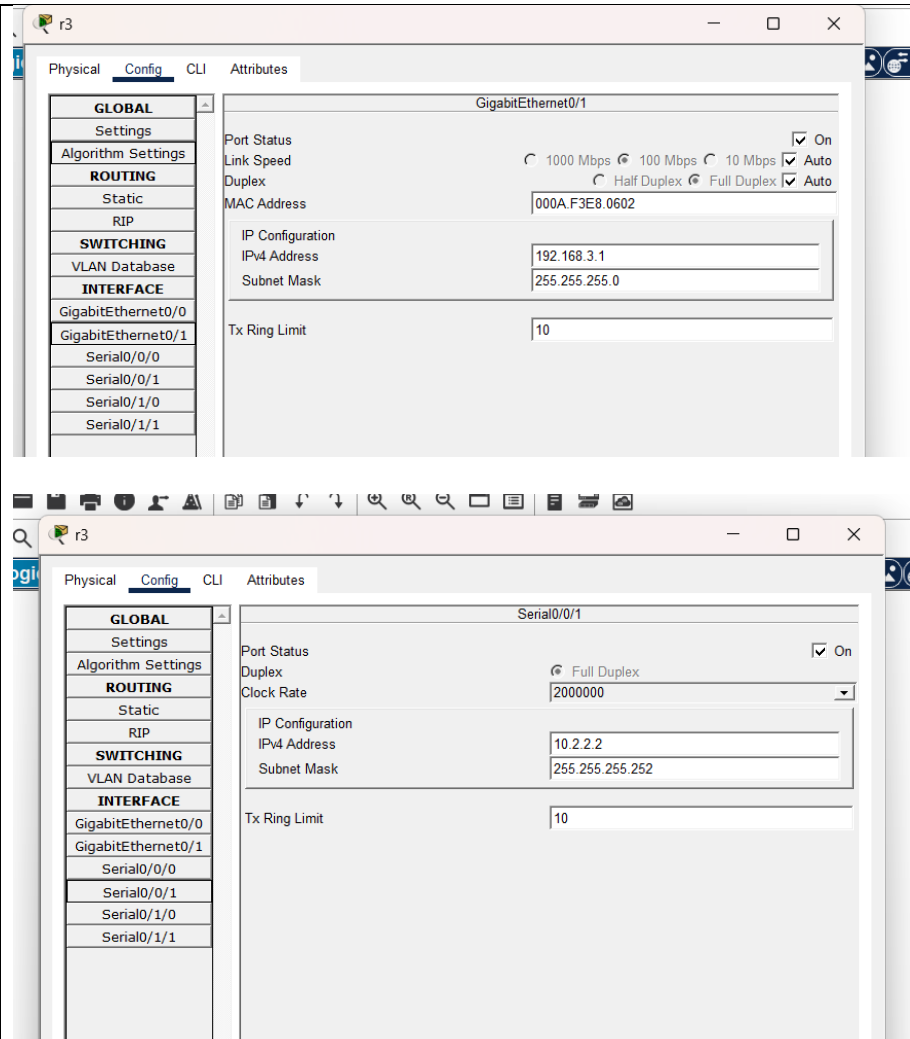


For Router 3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



Tactas server

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

tactas server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:97FF:FE13:17C3

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Radius server

radius server

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.3.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:4AFF:FE3B:C86D

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

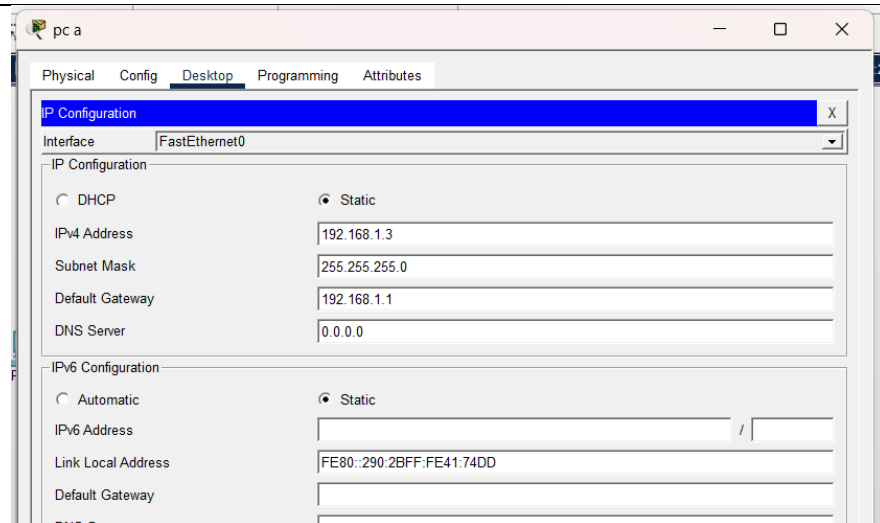
Password

PC A

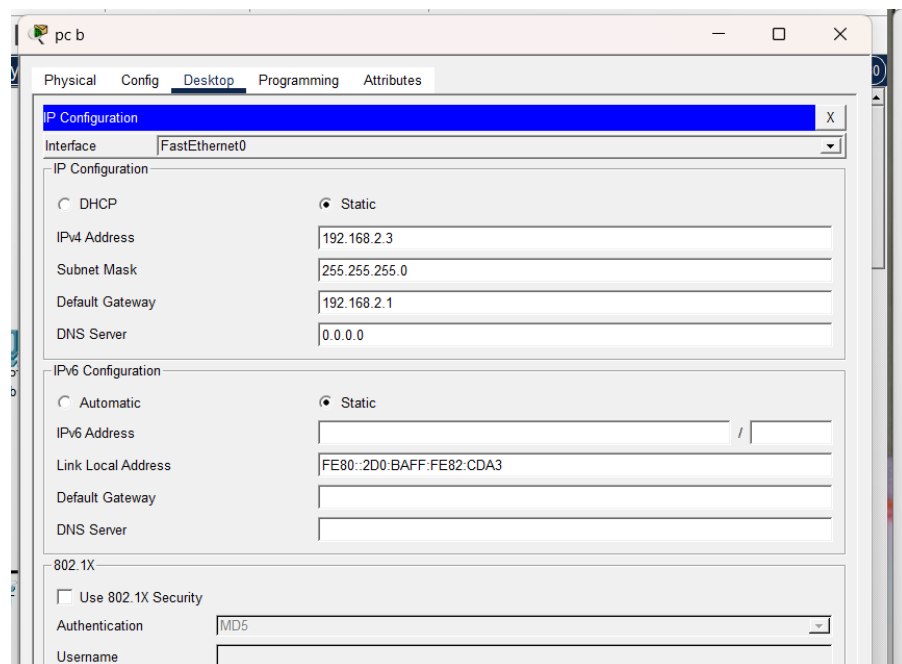
V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



PC B

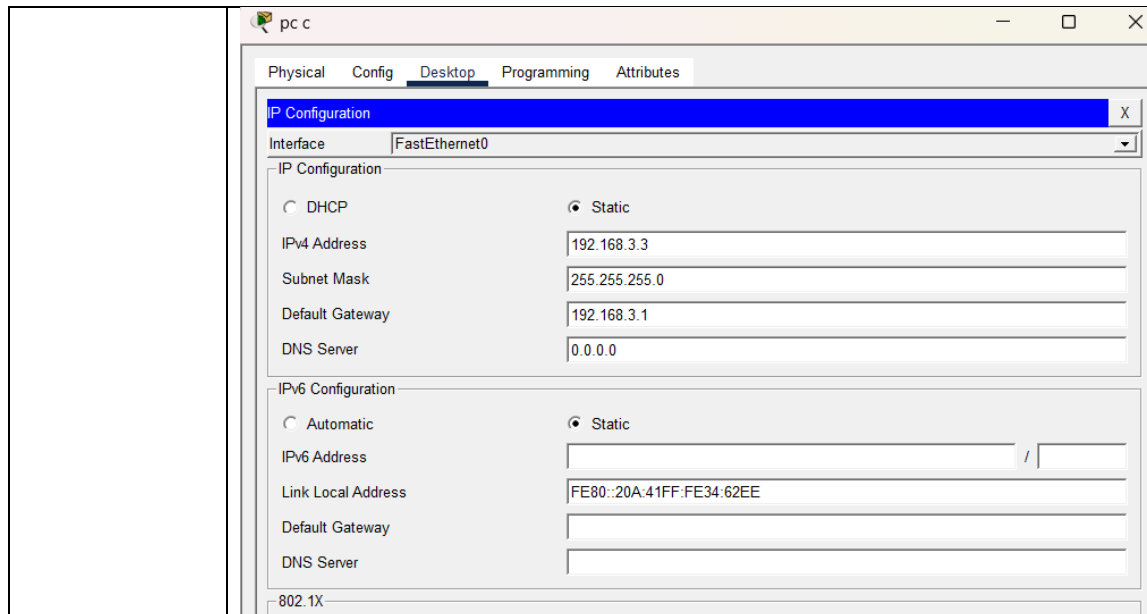


PC C

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



1) Bring up serial links (MOST IMPORTANT)

R1 (DCE on S0/0/0)

```
enable
conf t
int s0/0/0
clock rate 64000
no shutdown
end
```

R2 (DCE on S0/0/1)

```
enable
conf t
int s0/0/1
clock rate 64000
no shutdown
end
```

R2 and R3 (make sure other serial ports are up too)

```
conf t
int s0/0/0
no shutdown
int s0/0/1
no shutdown
end
```

(On R3 only s0/0/1 exists for this link—just do no shutdown there.)

After this, check:

```
show ip interface brief
You must see:
```

R1: s0/0/0 up up

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R2: s0/0/0 up up and s0/0/1 up up

R3: s0/0/1 up up

R1

```
Router(config)#interface Serial0/0/0
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#end
Router#
```

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       192.168.1.1     YES manual    up          up
Serial0/0/0              10.1.1.2        YES manual    up          up
Serial0/0/1              unassigned      YES unset    administratively down down
Serial0/1/0              unassigned      YES unset    administratively down down
Serial0/1/1              unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

R2

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s0/0/0
Router(config-if)#no shutdown
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int s0/0/1
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre> Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router ospf 1 Router(config-router)#network 192.168.2.0 0.0.0.255 area 0 Router(config-router)#network 10.1.1.0 0.0.0.3 area 0 Router(config-router)#network 10 02:56:40: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL, Loading Done % Invalid input detected at '^' marker. Router(config-router)#network 10.2.2.0 0.0.0.3 area 0 Router(config-router)#end Router# %SYS-5-CONFIG_I: Configured from console by console 02:57:50: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from LOADING to FULL, Loading Done Router#show ip ospf neighbor Neighbor ID Pri State Dead Time Address Interface 192.168.3.1 0 FULL/- 00:00:31 10.2.2.2 Serial0/0/1 192.168.1.1 0 FULL/- 00:00:32 10.1.1.2 Serial0/0/0 Router# </pre>
	R3 <pre> Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router ospf 1 Router(config-router)#network 192.168.3.0 0.0.0.255 area 0 Router(config-router)#network 10.2.2.0 0.0.0.3 area 0 Router(config-router)# 02:57:36: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial0/0/1 from LOADING to FULL, Loading Done end Router# %SYS-5-CONFIG_I: Configured from console by console </pre>
PART 1	Configure Local AAA Authentication for Console Access on R1
	Router 1 <pre> Router>enable Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#username Admin1 secret adminpa55 Router(config)#aaa new-model Router(config)#aaa authentication login default local Router(config)#line console 0 Router(config-line)#login authentication default Router(config-line)#end Router# %SYS-5-CONFIG_I: Configured from console by console exit </pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>User Access Verification Username: Admin1 Password: % Login invalid Username: Admin1 Password: Router></pre>
	PART 2 : Configure Local AAA Authentication for vty Lines on R1 <pre>R1#conf t Enter configuration commands, one per line. End with CNTL/Z. R1(config)#ip domain-name ccnasecurity.com R1(config)#crypto key generate rsa The name for the keys will be: R1.ccnasecurity.com Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes. How many bits in the modulus [512]: 1024 % Generating 1024 bit RSA keys, keys will be non-exportable...[OK] R1(config)#aaa authentication login SSH-LOGIN local *Mar 1 3:17:5.808: %SSH-5-ENABLED: SSH 1.99 has been enabled R1(config)#line vty 0 4 R1(config-line)#login authentication SSH-LOGIN R1(config-line)#transport input ssh R1(config-line)#end R1# %SYS-5-CONFIG_I: Configured from console by console</pre>
PART 3	Configure Server-Based AAA Authentication Using TACACS+ on R2 <pre>R2 (config)#username Admin2 secret admin2pa55 R2 (config)#exit Enter configuration commands, one per line. End with CNTL/ R2 (config)#aaa new-model R2 (config)#aaa authentication login default local R2 (config)#line vty 0 4 R2 (config-line)#login authentication default R2 (config-line)#exit R2 (config)#exit R2# %SYS-5-CONFIG_I: Configured from console by console R2>en R2#config Configuring from terminal, memory, or network [terminal]? Enter configuration commands, one per line. End with CNTL/Z. R2 (config)#tacacs-server host 192.168.2.2 R2 (config)#tacacs-server key tacacspa55 R2 (config)#aaa new-model R2 (config)#aaa authentication login default group tacacs+ local R2 (config)#line console 0 R2 (config-line)#login authentication default</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

PART 4

Configure Server-Based AAA Authentication using RADIUS on R3

```
Enter configuration commands, one per line. End with
Router(config)#username Admin3 secret admin3pa55
Router(config)#

Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#aaa new-model
Router(config)#radius-server host 192.168.3.2
Router(config)#radius-server key radiuspa55
Router(config)#aaa authentication login default group radius local
Router(config)#line console 0
Router(config-line)#login authentication default
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#exit
```

The screenshot shows a web-based configuration interface for a 'radius server'. The 'Services' tab is active, displaying the AAA configuration. The 'Service' is set to 'On' and the 'Radius Port' is 1645. Under 'Network Configuration', a table lists the configured RADIUS server:

	Client Name	Client IP	Server Type	Key
1	R3	192.168.3.1	Radius	radiuspa55

Below this, the 'User Setup' section shows a table of configured users:

	Username	Password
1	Admin3	admin3pa55

The interface includes 'Add', 'Save', and 'Remove' buttons for both the server and user tables. A 'Top' link is visible at the bottom left.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

V. G. VAZE KELKAR COLLEGE, MULUND (E)

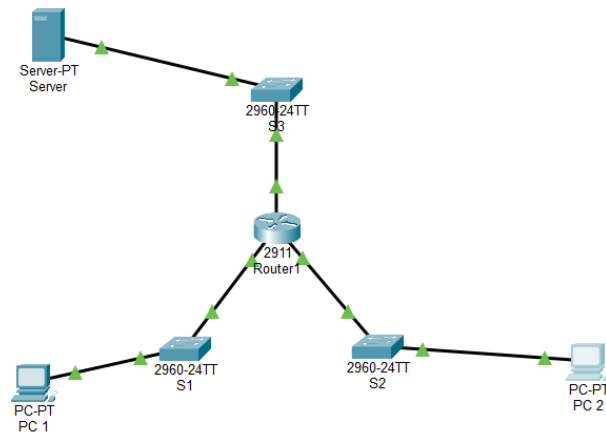
Roll No: A004

Name: Dhruv N Chandalia

PRACTICA L 3

Configuring Extended ACLs - Scenario 1

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.22.34.65	255.255.255.224	N/A
	G0/1	172.22.34.97	255.255.255.240	N/A
	G0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

Objectives

Part 1: Configure, Apply and Verify an Extended Numbered ACL

Part 2: Configure, Apply and Verify an Extended Named ACL

Background / Scenario

Two employees need access to services provided by the server. PC1 needs only FTP access while PC2 needs only web access. Both computers are able to ping the server, but not each other.

R1

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

The screenshot displays the Cisco Packet Tracer interface for configuring Router1. The 'Config' tab is active, showing the configuration for the GigabitEthernet0/0 interface. The left sidebar lists various configuration categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, and INTERFACE. Under the INTERFACE category, the GigabitEthernet0/0 interface is selected. The main configuration area shows the following settings for GigabitEthernet0/0:

- Port Status: ☒ On
- Link Speed: ☒ 1000 Mbps, ☒ 100 Mbps, ☒ 10 Mbps, ☒ Auto
- Duplex: ☒ Half Duplex, ☒ Full Duplex, ☒ Auto
- MAC Address: 0001.4212.BA01
- IP Configuration:
 - IPv4 Address: 172.22.34.65
 - Subnet Mask: 255.255.255.224
- Tx Ring Limit: 10

Below the configuration area, the 'Equivalent IOS Commands' section shows the following commands:

```
Router1
!
!
!
!
end

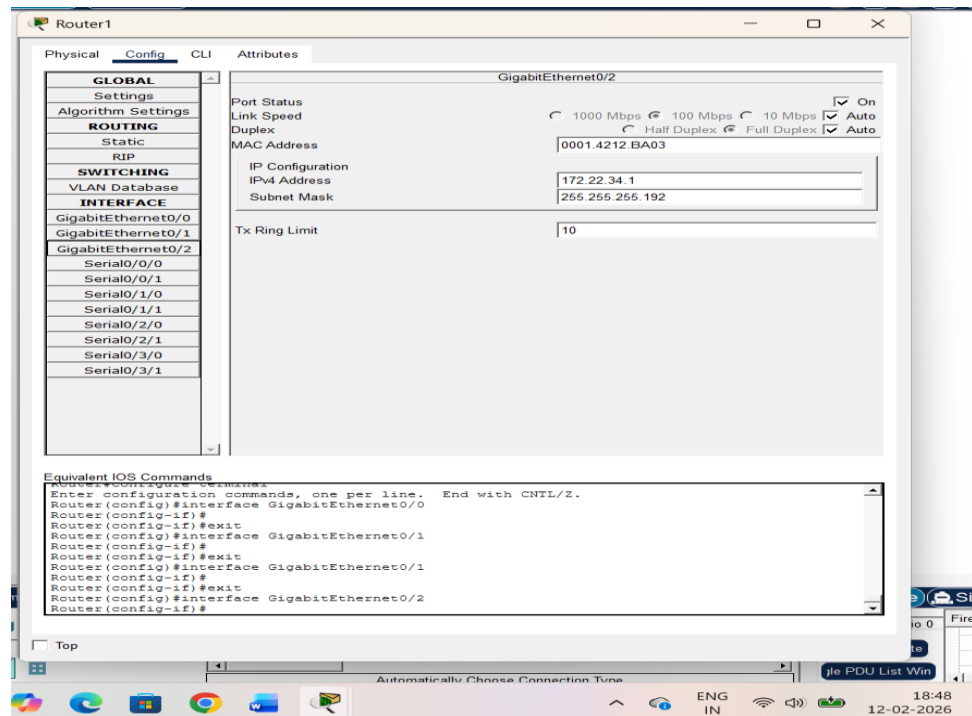
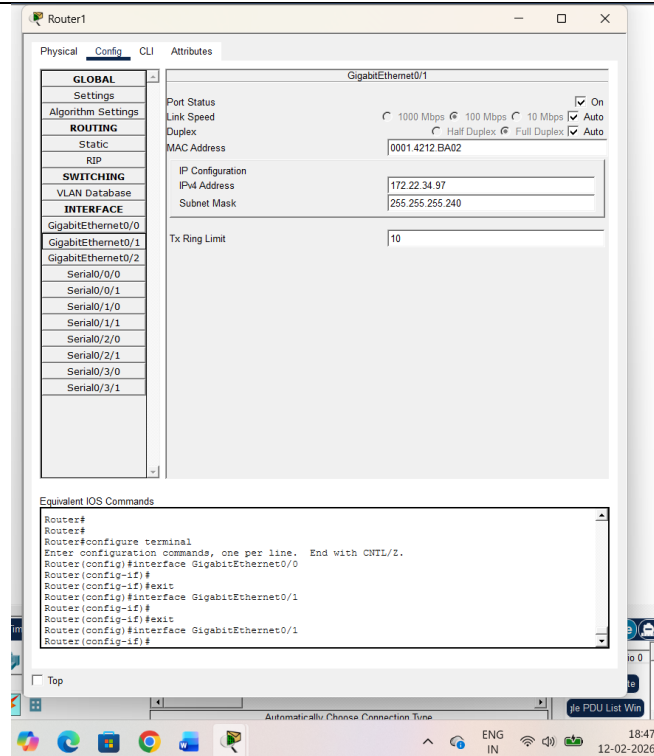
Router#
Router#
Router#configure terminal
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

The bottom of the window shows the Windows taskbar with the time 18:47 and date 12-02-2026.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

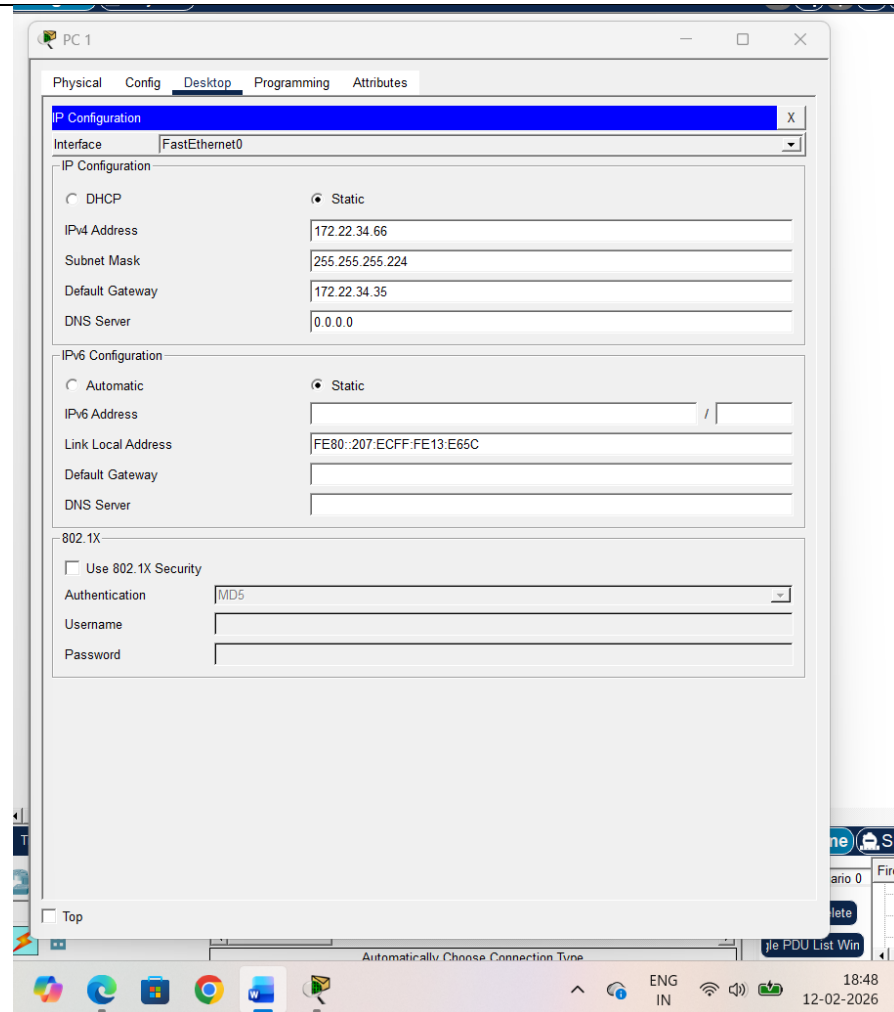


PC 1

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

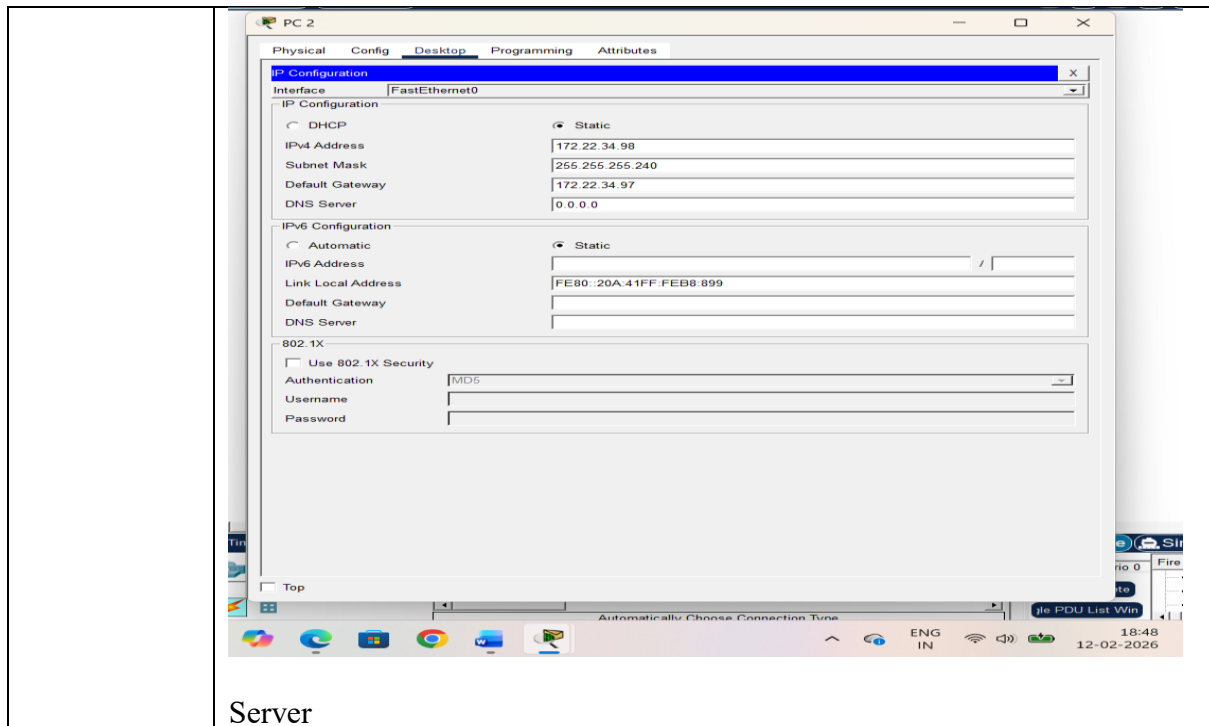


PC 2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

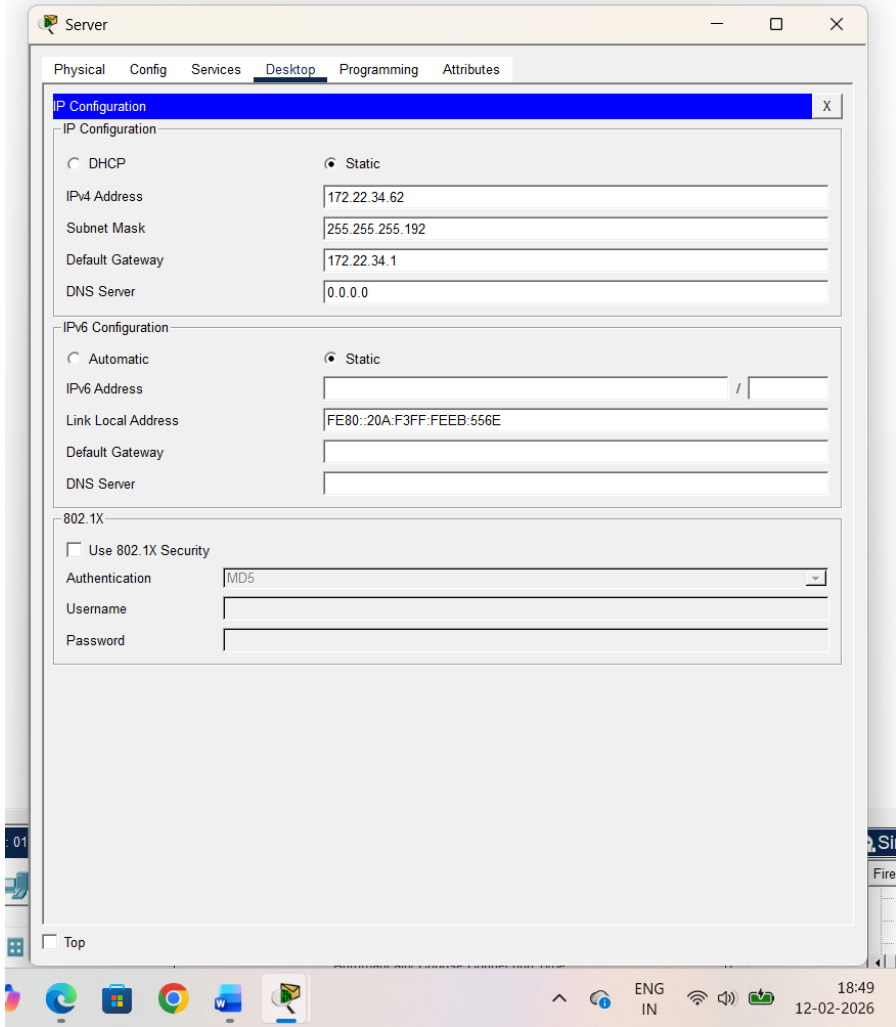
Name: Dhruv N Chandalia



V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	
Part 1	Configure, Apply and Verify an Extended Numbered ACL
Step 1	Configure an ACL to permit FTP and ICMP. - From global configuration mode on R1, enter the following command to determine the first valid number for an extended access list. - Add 100 to the command, followed by a question mark. - To permit FTP traffic, enter permit, followed by a question mark. - This ACL permits FTP and ICMP. ICMP is listed above, but FTP is not, because FTP uses TCP. Therefore, enter tcp to further refine the ACL help. - Notice that we could filter just for PC1 by using the host keyword or we could allow any host. In this case, any device is allowed that has an address belonging to the 172.22.34.64/27 network. Enter the network address, followed by a question mark. - Calculate the wildcard mask determining the binary opposite of a subnet mask. - Enter the wildcard mask, followed by a question mark.

Name: Dhruv N Chandalia

```

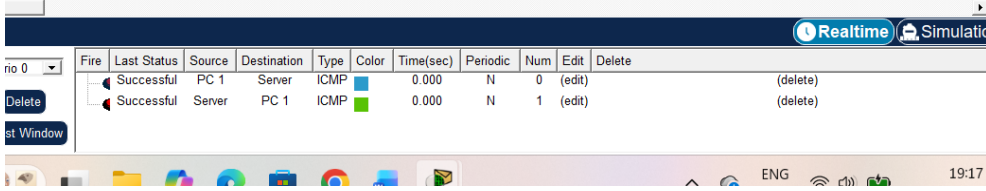
R1
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list ?
  <1-99>      IP standard access list
  <100-199>    IP extended access list
Router(config)#access-list 100 ?
  deny        Specify packets to reject
  permit      Specify packets to forward
  remark      Access list entry comment
Router(config)#access-list 100 permit ?
  ahp         Authentication Header Protocol
  eigrp       Cisco's EIGRP routing protocol
  esp         Encapsulation Security Payload
  gre         Cisco's GRE tunneling
  icmp        Internet Control Message Protocol
  ip          Any Internet Protocol
  ospf        OSPF routing protocol
  tcp         Transmission Control Protocol
  udp         User Datagram Protocol
Router(config)#access-list 100 permit tcp ?
  A.B.C.D     Source address
  any         Any source host
  host        A single source host
Router(config)#access-list 100 permit tcp 172.22.34.64 ?
  A.B.C.D     Source wildcard bits
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 ?
  A.B.C.D     Destination address
  any         Any destination host
  eq          Match only packets on a given port number
  gt          Match only packets with a greater port number
  host        A single destination host
  lt          Match only packets with a lower port number
  neq         Match only packets not on a given port number
  range       Match only packets in the range of port numbers
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 ?
  dscp        Match packets with given dscp value
  eq          Match only packets on a given port number
  established Match only packets with the established flag set
  gt          Match only packets with a greater port number
  lt          Match only packets with a lower port number
  neq         Match only packets not on a given port number
  precedence  Match packets with given precedence value
  range       Match only packets in the range of port numbers
  <cr>
Router(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq

```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

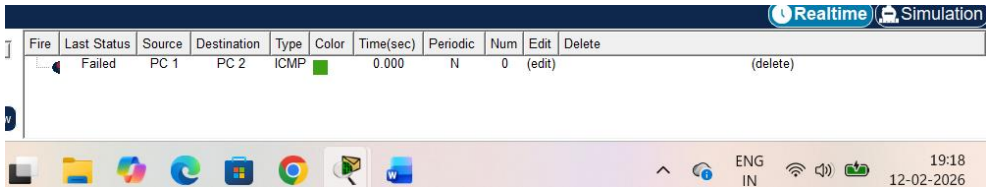
Name: Dhruv N Chandalia

	<pre>Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 100 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 100 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 100 permit % Incomplete command. Router(config)#access-list 100 permit icmp ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 100 permit icmp 172.22.34.64 ? A.B.C.D Source wildcard bits Router(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 ? A.B.C.D Destination address any Any destination host host A single destination host Router(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62 ? <0-256> type-num echo Echo (ping) echo-reply Echo reply host-unreachable Host unreachable net-unreachable Net unreachable port-unreachable Port unreachable protocol-unreachable Protocol unreachable ttl-exceeded TTL exceeded unreachable All unreachables <cr> Router(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62 Router(config)#</pre>
Step 2	Apply the ACL on the correct interface to filter traffic <pre><cr> Router(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62 Router(config)#interface gigabitEthernet 0/0 Router(config-if)#ip access-group 100 in Router(config-if)#</pre>
Step 3	Verify the ACL implementation 

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>ftp 172.22.34.62 Trying to connect...172.22.34.62 Connected to 172.22.34.62 220- Welcome to PT Ftp server Username:cisco 331- Username ok, need password Password:***** 230- Logged in (passive mode On) ftp>quit 221- Service closing control connection. C:\></pre> 
PART 2	Configure, Apply and Verify an Extended Named ACL
Step 1	Configure an ACL to permit HTTP access and ICMP. - Named ACLs start with the ip keyword. From global configuration mode of R1, enter the following command, followed by a question mark. - You can configure named standard and extended ACLs. This access list filters both source and destination IP addresses; therefore, it must be extended. Enter HTTP_ONLY as the name. (For Packet Tracer scoring, the name is case-sensitive.) - The prompt changes. You are now in extended named ACL configuration mode. All devices on the PC2 LAN need TCP access. Enter the network address, followed by a question mark. - Finish the statement by specifying the server address as you did in Part 1 and filtering www traffic.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

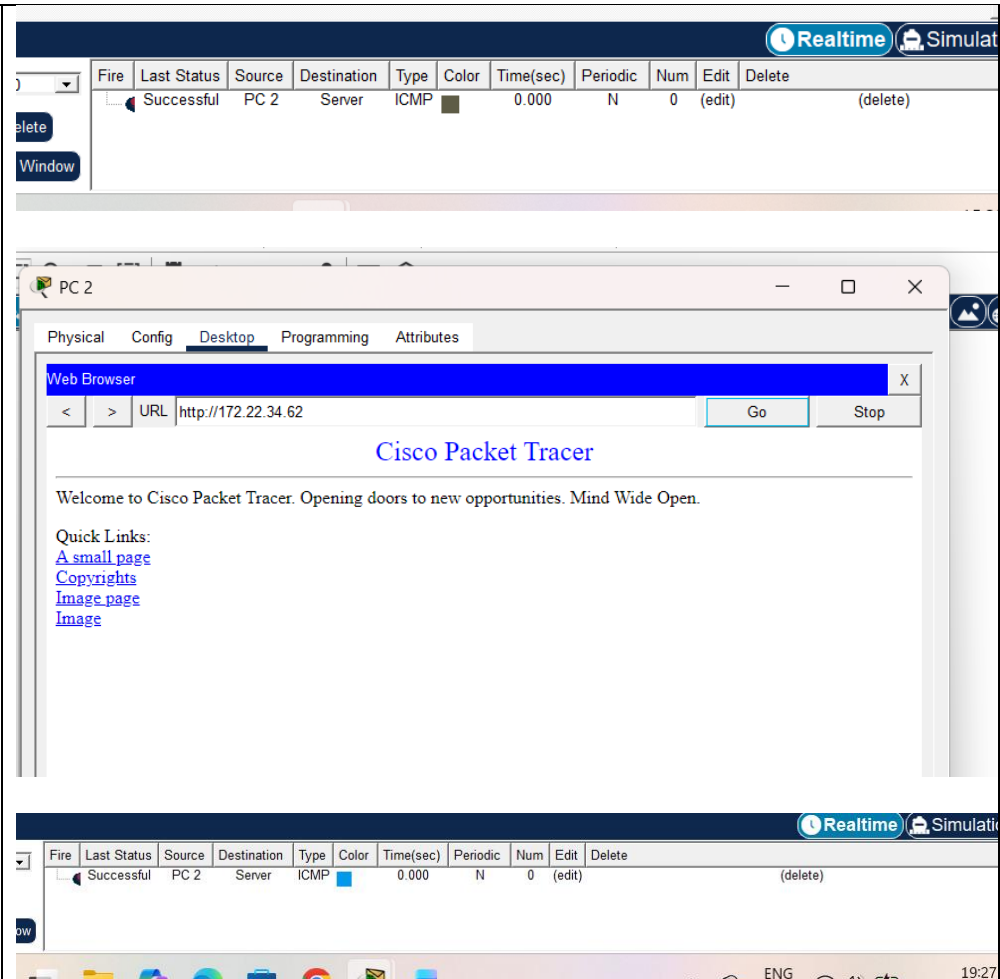
Name: Dhruv N Chandalia

	<pre>Router(config)#ip access-list ? extended Extended Access List standard Standard Access List Router(config)#ip access-list extended HTTP_ONLY Router(config-ext-nacl)#permit tcp 172.22.34.96 ? A.B.C.D Source wildcard bits Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 ? A.B.C.D Destination address any Any destination host eq Match only packets on a given port number gt Match only packets with a greater port number host A single destination host lt Match only packets with a lower port number neq Match only packets not on a given port number range Match only packets in the range of port numbers Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 ? eq Match only packets on a given port number established established gt Match only packets with a greater port number lt Match only packets with a lower port number neq Match only packets not on a given port number range Match only packets in the range of port numbers <cr> Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq ? <0-65535> Port number domain Domain Name Service (DNS, 53) ftp File Transfer Protocol (21) pop3 Post Office Protocol v3 (110) smtp Simple Mail Transport Protocol (25) telnet Telnet (23) www World Wide Web (HTTP, 80) Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www Router(config-ext-nacl)# % incomplete command. Router(config-ext-nacl)#permit tcp ? A.B.C.D Source address any Any source host host A single source host Router(config-ext-nacl)#permit tcp 172.22.34.96 ? A.B.C.D Source wildcard bits Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15? A.B.C.D Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15host 172.22.34.62 Router(config)#interface gigabitEthernet 0/1 Router(config-if)#ip access_group HTTP_ONLY in ^ % Invalid input detected at '^' marker. Router(config-if)#ip access-group HTTP_ONLY in</pre>
	Verify <pre>C:\>ftp 172.22.34.66 Trying to connect...172.22.34.66 %Error opening ftp://172.22.34.66/ (Timed out) . (Disconnecting from ftp server)</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

V. G. VAZE KELKAR COLLEGE, MULUND (E)

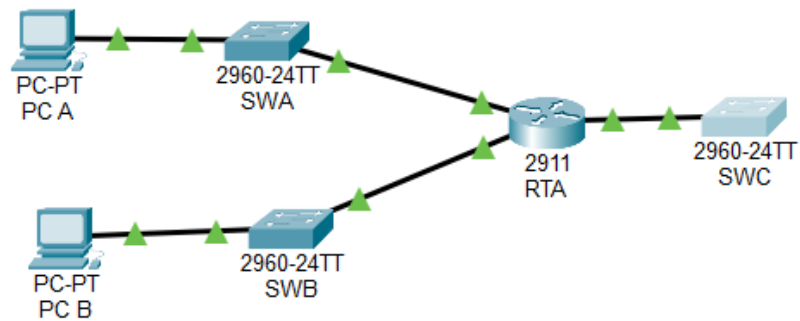
Roll No: A004

Name: Dhruv N Chandalia

PRACTICA L 3

Configuring Extended ACLs - Scenario 2

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
RTA	G0/0	10.101.117.49	255.255.255.248	N/A
	G0/1	10.101.117.33	255.255.255.240	N/A
	G0/2	10.101.117.1	255.255.255.224	N/A
PCA	NIC	10.101.117.51	255.255.255.248	10.101.117.49
PCB	NIC	10.101.117.35	255.255.255.240	10.101.117.33
SWA	VLAN 1	10.101.117.50	255.255.255.248	10.101.117.49
SWB	VLAN 1	10.101.117.34	255.255.255.240	10.101.117.33
SWC	VLAN 1	10.101.117.2	255.255.255.224	10.101.117.1

Objectives

Part 1: Configure, Apply and Verify an Extended Numbered ACL

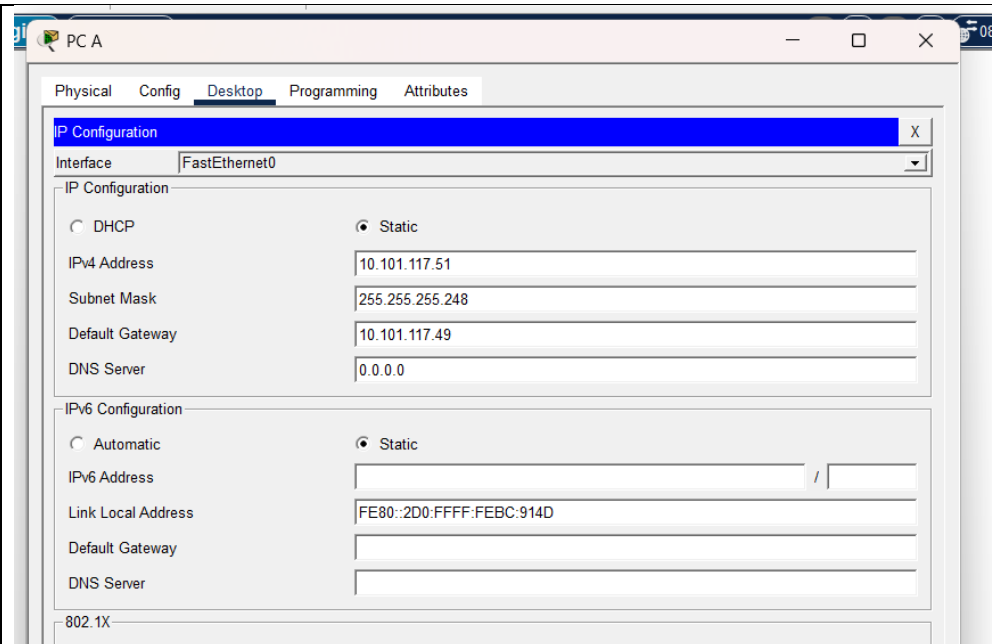
Part 2: Reflection Questions

PC A

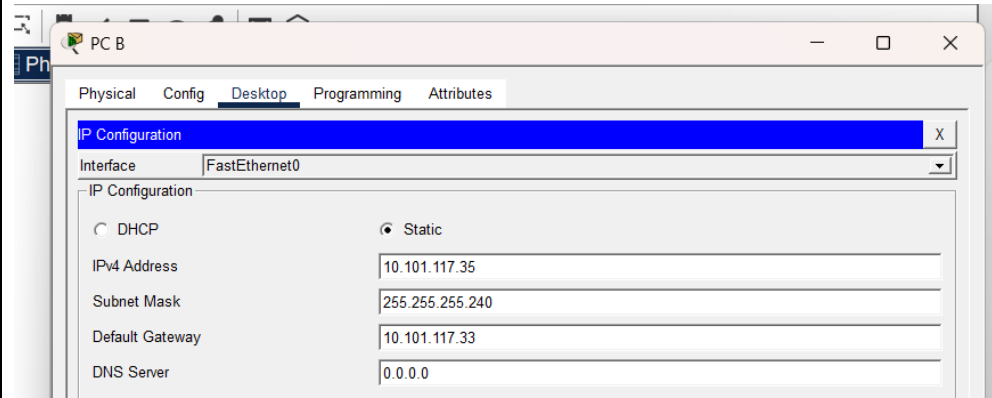
V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



PC B



RTA

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

RTA

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/0

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.F700.8D01

IP Configuration

IPv4 Address 10.101.117.49

Subnet Mask 255.255.255.248

Tx Ring Limit 10

RTA

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.F700.8D02

IP Configuration

IPv4 Address 10.101.117.33

Subnet Mask 255.255.255.240

Tx Ring Limit 10

RTA

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 00E0.F700.8D03

IP Configuration

IPv4 Address 10.101.117.1

Subnet Mask 255.255.255.224

Tx Ring Limit 10

SWA

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>Switch>enable Switch# Switch#configure terminal Enter configuration commands, one per line. End with CNTL/Z. Switch(config)# Switch(config)#interface vlan 1 Switch(config-if)#ip address 10.101.117.50 255.255.255.248 Switch(config-if)#no shutdown Switch(config-if)# %LINK-3-UPDOWN: Interface Vlan1, changed state to down Switch(config-if)#ip default-gateway 10.101.117.49 Switch(config)#</pre> SWB <pre>Switch>en Switch#conf t Enter configuration commands, one per line. End with CNTL/Z. Switch(config)#interface vlan 1 Switch(config-if)#ip address 10.101.117.34 255.255.255.240 Switch(config-if)#no shutdown Switch(config-if)# %LINK-3-UPDOWN: Interface Vlan1, changed state to down Switch(config-if)#ip default-gateway 10.101.117.33 Switch(config)#</pre> SWC <pre>Switch>en Switch#conf t Enter configuration commands, one per line. End with CNTL/Z. Switch(config)#interface vlan 1 Switch(config-if)#ip address 10.101.117.2 255.255.255.224 Switch(config-if)#no shutdown Switch(config-if)# %LINK-3-UPDOWN: Interface Vlan1, changed state to down Switch(config-if)#ip default-gateway 10.101.117.1 Switch(config)#exit Switch# %SYS-5-CONFIG_I: Configured from console by console</pre>
Step 1	Configure the extended ACL. RTA

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access list ?
% Unrecognized command
Router(config)#access-list ?
<1-99> IP standard access list
<100-199> IP extended access list
Router(config)#access-list 199 ?
deny Specify packets to reject
permit Specify packets to forward
remark Access list entry comment
Router(config)#access-list 199 permit ?
ahp Authentication Header Protocol
eigrp Cisco's EIGRP routing protocol
esp Encapsulation Security Payload
gre Cisco's GRE tunneling
icmp Internet Control Message Protocol
ip Any Internet Protocol
ospf OSPF routing protocol
tcp Transmission Control Protocol
udp User Datagram Protocol
Router(config)#access-list 199 permit tcp ?
A.B.C.D Source address
any Any source host
host A single source host
Router(config)#access-list 199 permit tcp 10.101.117.32 ?
A.B.C.D Source wildcard bits
Router(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 ?
A.B.C.D Destination address
any Any destination host
eq Match only packets on a given port number
gt Match only packets with a greater port number
host A single destination host
lt Match only packets with a lower port number
neq Match only packets not on a given port number
range Match only packets in the range of port numbers
Router(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 ?
A.B.C.D Destination wildcard bits
Router(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 ?
dscp Match packets with given dscp value
eq Match only packets on a given port number
established established
gt Match only packets with a greater port number
lt Match only packets with a lower port number
neq Match only packets not on a given port number
precedence Match packets with given precedence value
range Match only packets in the range of port numbers
<cr>

neq Match only packets not on a given port number
precedence Match packets with given precedence value
range Match only packets in the range of port numbers
<cr>
Router(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq
% Incomplete command.
Router(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22
Router(config)#access-list?
access-list
Router(config)#access-list ?
<1-99> IP standard access list
<100-199> IP extended access list
Router(config)#access-list 199 ?
deny Specify packets to reject
permit Specify packets to forward
remark Access list entry comment
Router(config)#access-list 199 permit ?
ahp Authentication Header Protocol
eigrp Cisco's EIGRP routing protocol
esp Encapsulation Security Payload
gre Cisco's GRE tunneling
icmp Internet Control Message Protocol
ip Any Internet Protocol
ospf OSPF routing protocol
tcp Transmission Control Protocol
udp User Datagram Protocol
Router(config)#access-list 199 permit icmp ?
A.B.C.D Source address
any Any source host
host A single source host
Router(config)#access-list 199 permit icmp any ?
A.B.C.D Destination address
any Any destination host
host A single destination host
Router(config)#access-list 199 permit icmp any any
Router(config)#
```

Copy

Paste

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

Step 2	<pre>Router(config)#interface g0/2 Router(config-if)#ip access-group 199 out Router(config-if)#</pre> apply extended ACL																																																																													
Step 3	Verification a RealtimeSimulation <table><tr><th>Fire</th><th>Last Status</th><th>Source</th><th>Destination</th><th>Type</th><th>Color</th><th>Time(sec)</th><th>Periodic</th><th>Num</th><th>Edit</th><th>Delete</th></tr><tr><td></td><td>Successful</td><td>PC B ...</td><td>PC A /29</td><td>ICMP</td><td></td><td>0.000</td><td>N</td><td>0</td><td>(edit)</td><td>(delete)</td></tr><tr><td></td><td>Successful</td><td>PC B ...</td><td>SWA</td><td>ICMP</td><td></td><td>0.000</td><td>N</td><td>1</td><td>(edit)</td><td>(delete)</td></tr><tr><td></td><td>Successful</td><td>PC B ...</td><td>RTA</td><td>ICMP</td><td></td><td>0.000</td><td>N</td><td>2</td><td>(edit)</td><td>(delete)</td></tr><tr><td></td><td>Successful</td><td>PC B ...</td><td>SWC /27</td><td>ICMP</td><td></td><td>0.000</td><td>N</td><td>3</td><td>(edit)</td><td>(delete)</td></tr></table> ENG16:54 RealtimeSimulation <table><tr><th>Fire</th><th>Last Status</th><th>Source</th><th>Destination</th><th>Type</th><th>Color</th><th>Time(sec)</th><th>Periodic</th><th>Num</th><th>Edit</th><th>Delete</th></tr><tr><td></td><td>Successful</td><td>PC B ...</td><td>SWB</td><td>ICMP</td><td></td><td>0.000</td><td>N</td><td>0</td><td>(edit)</td><td>(delete)</td></tr></table> ENG16:5 B b. SSH from PCB to SWC. The username is Admin, and the password is Adminpa55. On SWC Generate RSA <pre>SWC# SWC#conf t Enter configuration commands, one per line. End with CNTL/Z. SWC(config)#crypto key generate rsa The name for the keys will be: SWC.lab.local Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes. How many bits in the modulus [512]: 1024 % Generating 1024 bit RSA keys, keys will be non-exportable...[OK] SWC(config)#end *Mar 1 1:41:1.774: %SSH-5-ENABLED: SSH 2 has been enabled SWC# %SYS-5-CONFIG I: Configured from console by console</pre>	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete		Successful	PC B ...	PC A /29	ICMP		0.000	N	0	(edit)	(delete)		Successful	PC B ...	SWA	ICMP		0.000	N	1	(edit)	(delete)		Successful	PC B ...	RTA	ICMP		0.000	N	2	(edit)	(delete)		Successful	PC B ...	SWC /27	ICMP		0.000	N	3	(edit)	(delete)	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete		Successful	PC B ...	SWB	ICMP		0.000	N	0	(edit)	(delete)
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete																																																																				
	Successful	PC B ...	PC A /29	ICMP		0.000	N	0	(edit)	(delete)																																																																				
	Successful	PC B ...	SWA	ICMP		0.000	N	1	(edit)	(delete)																																																																				
	Successful	PC B ...	RTA	ICMP		0.000	N	2	(edit)	(delete)																																																																				
	Successful	PC B ...	SWC /27	ICMP		0.000	N	3	(edit)	(delete)																																																																				
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete																																																																				
	Successful	PC B ...	SWB	ICMP		0.000	N	0	(edit)	(delete)																																																																				

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#hostname SWC
SWC(config)#
SWC(config)#interface vlan 1
SWC(config-if)#ip address 10.101.117.2 255.255.255.224
SWC(config-if)#no shutdown
SWC(config-if)#exit
SWC(config)#
SWC(config)#ip default-gateway 10.101.117.1
SWC(config)#
SWC(config)#username Admin secret Adminpa55
SWC(config)#ip domain-name lab.local
SWC(config)#crypto key generate rsa modulus 1024
^
% Invalid input detected at '^' marker.

SWC(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
SWC(config)#
SWC(config)#line vty 0 15
SWC(config-line)#login local
SWC(config-line)#transport input ssh
SWC(config-line)#end
SWC#wr
Building configuration...
[OK]
SWC#
%SYS-5-CONFIG_I: Configured from console by console

SWC#
```

Then on PC B

```
C:\>ssh -l Admin 10.101.117.2

Password:
% Login invalid

Password:
% Login invalid

Password:

SWC>
```

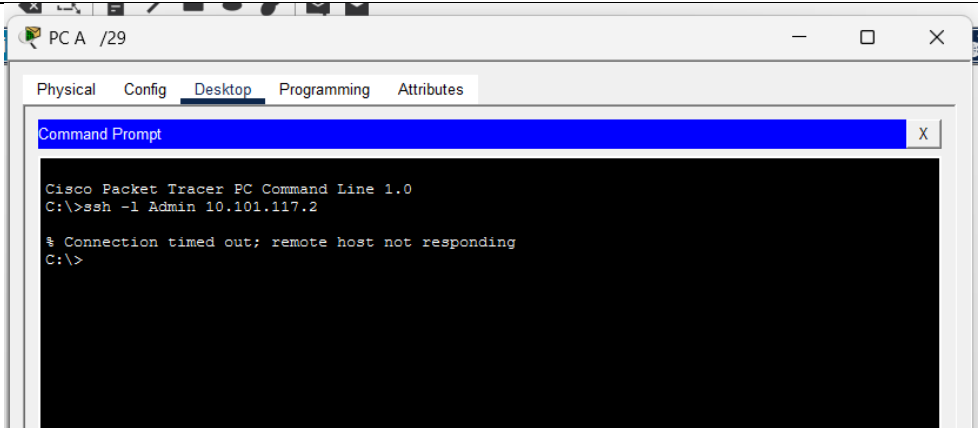
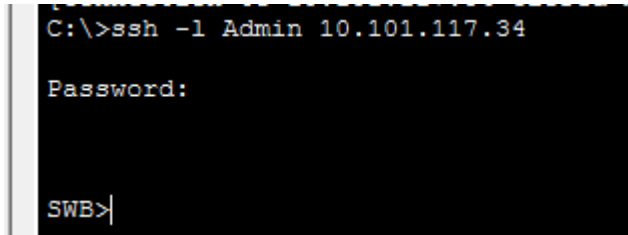
C

Realtime Simulation										
	Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit Delete
		Successful	PCA ...	SWB	ICMP		0.000	N	0	(edit) (delete)
		Successful	PCA ...	PC B /28	ICMP		0.000	N	1	(edit) (delete)
		Successful	PCA ...	RTA	ICMP		0.000	N	2	(edit) (delete)
		Successful	PCA ...	SWC /27	ICMP		0.000	N	3	(edit) (delete)

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	
Scenario 2	Step 1: SSH from PCA to SWB. The access list is placed on G0/2 and does not affect this connection. The username is Admin, and the password is Adminpa55. <pre>Switch>en Switch#conf t Enter configuration commands, one per line. End with CNTL/Z. Switch(config)#hostname SWB SWB(config)#interface vlan 1 SWB(config-if)#ip address 10.101.117.34 255.255.255.240 SWB(config-if)#no shutdown SWB(config-if)#exit SWB(config)#ip default-gateway 10.101.117.33 SWB(config)#username Admin secret Adminpa55 SWB(config)#ip domain-name lab.local SWB(config)#crypto key generate rsa The name for the keys will be: SWB.lab.local Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes. How many bits in the modulus [512]: 1024 % Generating 1024 bit RSA keys, keys will be non-exportable...[OK] SWB(config)#ip ssh version 2 *Mar 1 2:24:20.240: %SSH-5-ENABLED: SSH 1.99 has been enabled SWB(config)#line vty 0 15 SWB(config-line)#login local SWB(config-line)#transport input ssh SWB(config-line)#end SWB# %SYS-5-CONFIG_I: Configured from console by console SWB#wr Building configuration... [OK] SWB#</pre> 

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>SWB#ssh -l Admin 10.101.117.2 Password: % Login invalid Password: SWC></pre>																																					
Part 2	Step 2: Reflection Questions																																					
	1. How was PCA able to bypass access list 199 and SSH to SWC? Two steps were used: First, PCA used SSH to access SWB. From SWB, SSH was allowed to SWC. 2. What could have been done to prevent PCA from accessing SWC indirectly, while allowing PCB SSH access to SWC? Because it was requested to block all traffic to 10.101.117.0/27 except SSH traffic originating from 10.101.117.32/28 the access list could be written as is. Instead of applying the ACL to G0/2 outbound apply the same ACL to both G0/0 and G0/1 inbound. Suggested Scoring Rubric <table><tr><th>Activity Section</th><th>Question Location</th><th>Possible Points</th><th>Earned Points</th></tr><tr><td rowspan="3">Part 1: Configure, Apply and Verify an Extended Numbered ACL</td><td>Step 1a</td><td>4</td><td></td></tr><tr><td>Step 1b</td><td>4</td><td></td></tr><tr><td>Step 2</td><td>4</td><td></td></tr><tr><td colspan="2">Part 1 Total</td><td>12</td><td></td></tr><tr><td rowspan="2">Part 2: Reflection Questions</td><td>Question 1</td><td>4</td><td></td></tr><tr><td>Question 2</td><td>4</td><td></td></tr><tr><td colspan="2">Part 2 Total</td><td>8</td><td></td></tr><tr><td colspan="2">Packet Tracer Score</td><td>80</td><td></td></tr><tr><td colspan="2">Total Score</td><td>100</td><td></td></tr></table>	Activity Section	Question Location	Possible Points	Earned Points	Part 1: Configure, Apply and Verify an Extended Numbered ACL	Step 1a	4		Step 1b	4		Step 2	4		Part 1 Total		12		Part 2: Reflection Questions	Question 1	4		Question 2	4		Part 2 Total		8		Packet Tracer Score		80		Total Score		100	
Activity Section	Question Location	Possible Points	Earned Points																																			
Part 1: Configure, Apply and Verify an Extended Numbered ACL	Step 1a	4																																				
	Step 1b	4																																				
	Step 2	4																																				
Part 1 Total		12																																				
Part 2: Reflection Questions	Question 1	4																																				
	Question 2	4																																				
Part 2 Total		8																																				
Packet Tracer Score		80																																				
Total Score		100																																				

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R2

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface loopback 0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

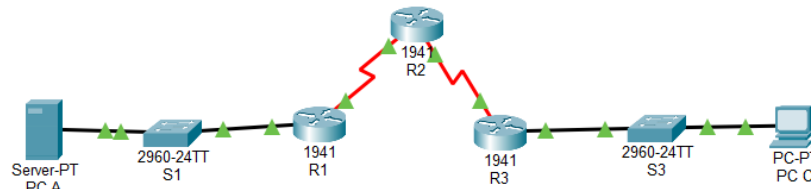
Roll No: A004

Name: Dhruv N Chandalia

PRACTICAL
L 4

Configure IP ACLs to Mitigate Attacks.

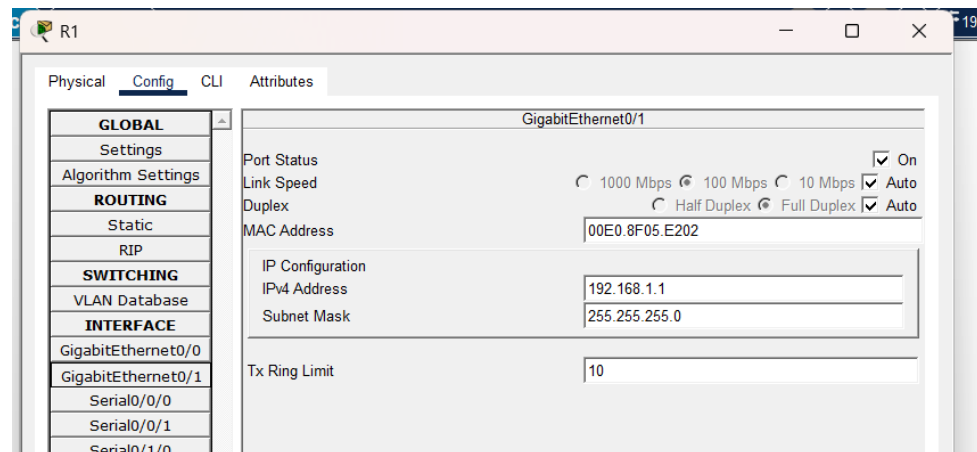
Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	Lo0	192.168.2.1	255.255.255.0	N/A	N/A
	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

R1



V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.1.1.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

R2

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.1.1.2

Subnet Mask 255.255.255.252

Tx Ring Limit 10

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

Serial0/0/1

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.2.2.2

Subnet Mask 255.255.255.252

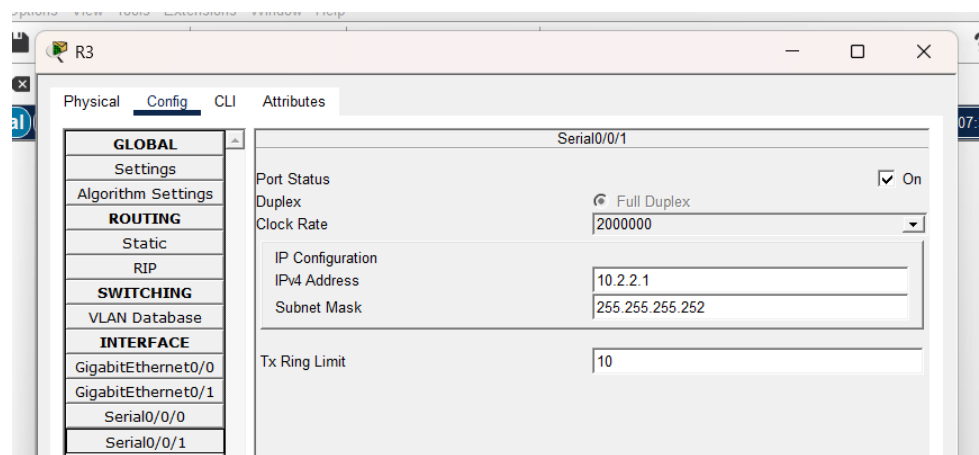
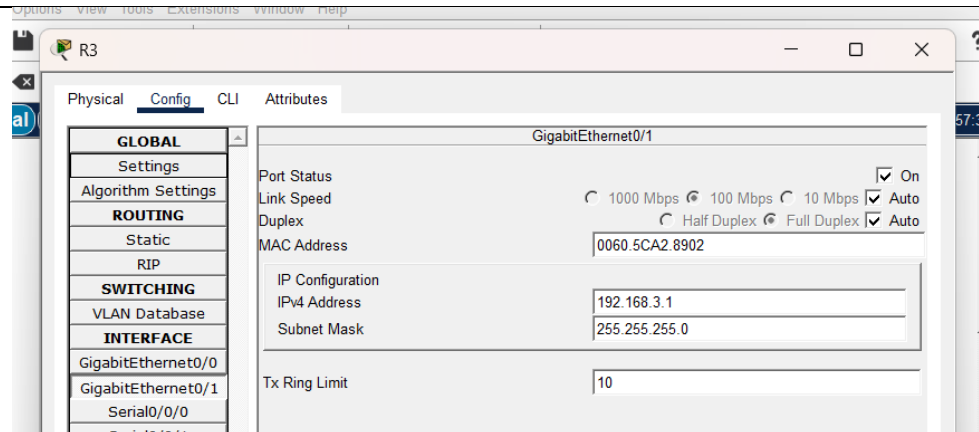
Tx Ring Limit 10

R3

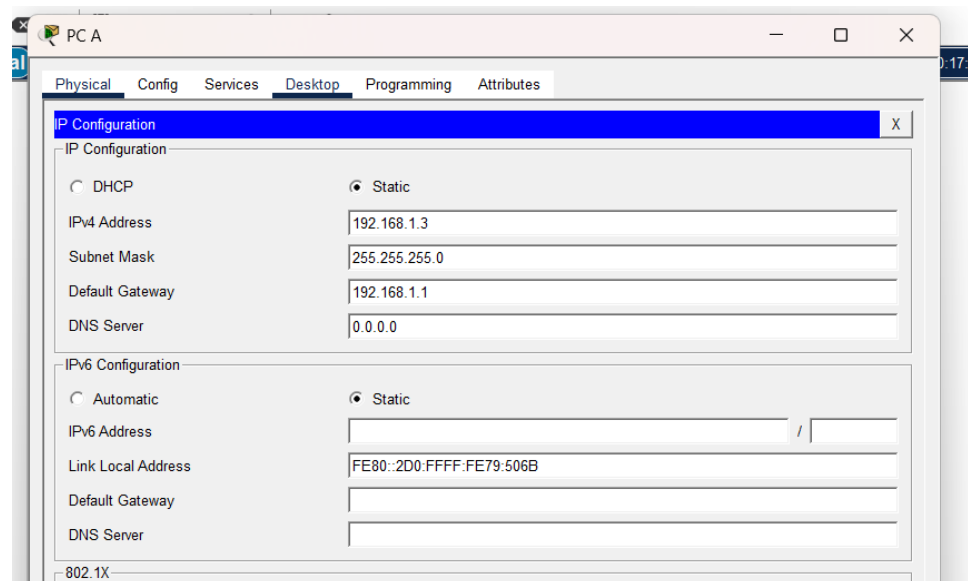
V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



PC A

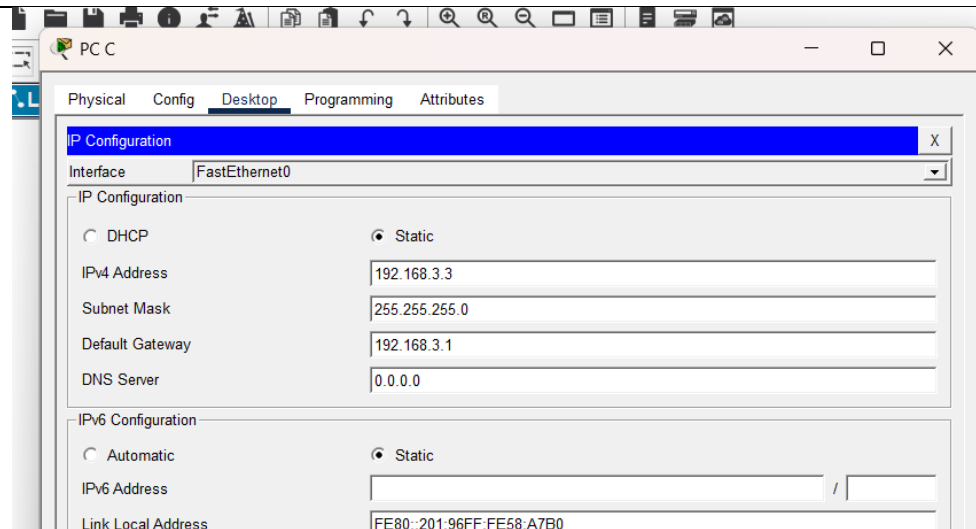


PC C

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



Add 2 static routes on R2

On R2, type:

enable

conf t

```
ip route 192.168.1.0 255.255.255.0 10.1.1.1
```

```
ip route 192.168.3.0 255.255.255.0 10.2.2.1
```

end

wr

Meaning:

- To reach 192.168.1.x → send to R1 (10.1.1.1)
- To reach 192.168.3.x → send to R3 (10.2.2.1)

PART 1

Verify Basic Network Connectivity

STEP 1

From PC-A, verify connectivity to PC-C and R2.

From the command prompt, ping PC-C (192.168.3.3).

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=12ms TTL=125
Reply from 192.168.3.3: bytes=32 time=8ms TTL=125
Reply from 192.168.3.3: bytes=32 time=2ms TTL=125
Reply from 192.168.3.3: bytes=32 time=8ms TTL=125

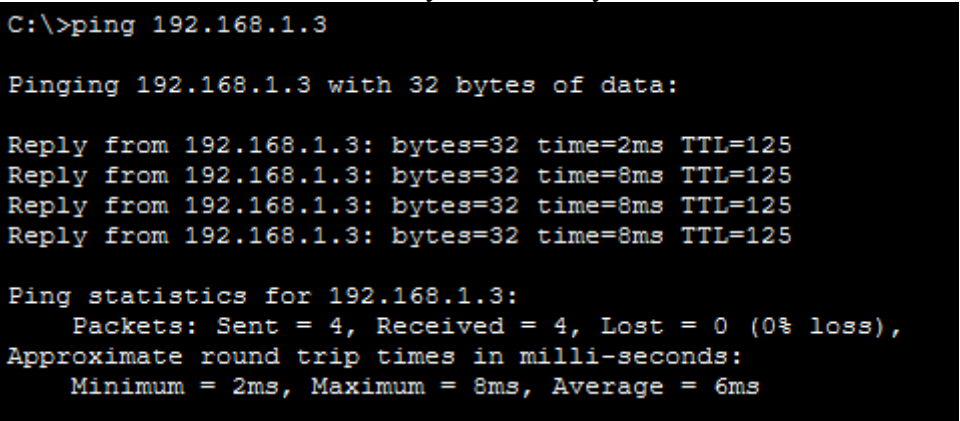
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 7ms

C:\>
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	From the command prompt, establish an SSH session to R2 Lo0 interface (192.168.2.1) using username SSHadmin and password ciscosshpa55. When finished, exit the SSH session. SERVER> ssh -l SSHadmin 192.168.2.1 <pre>----- Router(config)#hostname R2 R2(config)#ip domain-name lab.local R2(config)#username SSHadmin secret cisco R2(config)#crypto key generate rsa The name for the keys will be: R2.lab.local Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes. How many bits in the modulus [512]: 1024 % Generating 1024 bit RSA keys, keys will be non-exportable...[OK] R2(config)#ip ssh version 2 *Mar 2 17:42:0.945: %SSH-5-ENABLED: SSH 1.99 has been enabled R2(config)#line vty 0 4 R2(config-line)#login local R2(config-line)#transport input ssh R2(config-line)#end R2# %SYS-5-CONFIG_I: Configured from console by console wr Building configuration... [OK]</pre>
STEP 2	a. From PC-C, verify connectivity to PC-A and R2.  <pre>C:\>ping 192.168.1.3 Pinging 192.168.1.3 with 32 bytes of data: Reply from 192.168.1.3: bytes=32 time=2ms TTL=125 Reply from 192.168.1.3: bytes=32 time=8ms TTL=125 Reply from 192.168.1.3: bytes=32 time=8ms TTL=125 Reply from 192.168.1.3: bytes=32 time=8ms TTL=125 Ping statistics for 192.168.1.3: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 8ms, Average = 6ms</pre> b. From the command prompt, establish an SSH session to R2 Lo0 interface (192.168.2.1) using username SSHadmin and password ciscosshpa55. Close the SSH session when finished. PC> ssh -l SSHadmin 192.168.2.1 On R3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.2.0 255.255.255.0 10.2.2.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
Router#show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
Known via "static", distance 1, metric 0
Routing Descriptor Blocks:
* 10.2.2.2
  Route metric is 0, traffic share count is 1

Router#
```

On PC C

```
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=9ms TTL=254
Reply from 192.168.2.1: bytes=32 time=7ms TTL=254
Reply from 192.168.2.1: bytes=32 time=7ms TTL=254
Reply from 192.168.2.1: bytes=32 time=7ms TTL=254

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 9ms, Average = 7ms

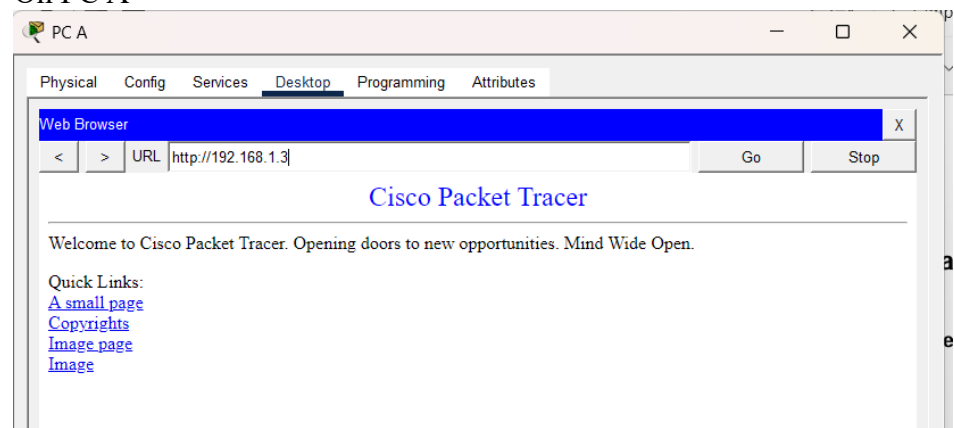
C:\>ssh -l SSHAdmin 192.168.2.1

Password:
% Login invalid

Password:

R2>
```

On PC A



V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

PART 2	Secure Access to Routers
Step 1	Configure ACL 10 to block all remote access to the routers except from PC-C. Use the access-list command to create a numbered IP ACL on R1, R2, and R3. On R1 <pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 10 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 10 permit ? A.B.C.D Address to match any Any source host host A single host address Router(config)#access-list 10 permit host 192.168.3.3 Router(config)#</pre> On R2 <pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 10 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 10 permit ? A.B.C.D Address to match any Any source host host A single host address Router(config)#access-list 10 permit host 192.168.3.3 Router(config)#</pre> On R3 <pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 10 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 10 permit ? A.B.C.D Address to match any Any source host host A single host address Router(config)#access-list 10 permit host 192.168.3.3 Router(config)#</pre>
Step 2	Apply ACL 10 to ingress traffic on the VTY lines. Use the access-class command to apply the access list to incoming traffic on the VTY lines.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

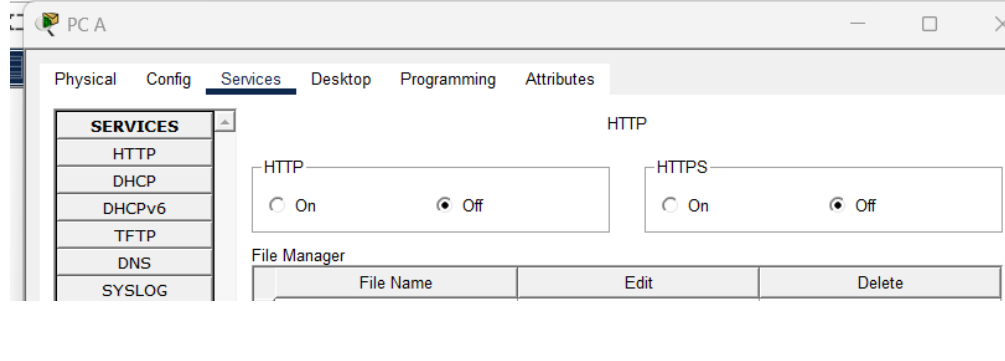
Name: Dhruv N Chandalia

	On R1 <pre>Router(config)#line vty 0 4 Router(config-line)#access-class 10 in Router(config-line)#end Router# %SYS-5-CONFIG_I: Configured from console by console wr Building configuration... [OK] Router#</pre> On R2 <pre>Router(config)#line vty 0 4 Router(config-line)#access-class 10 in Router(config-line)#end Router# %SYS-5-CONFIG_I: Configured from console by console wr Building configuration... [OK] Router#</pre> On R3 <pre>Router(config)#line vty 0 4 Router(config-line)#access-class 10 in Router(config-line)#end Router# %SYS-5-CONFIG_I: Configured from console by console wr Building configuration... [OK] Router#</pre>
Step 3	Verify exclusive access from management station PC-C.
	On PC C <pre>C:\>ssh -l SSHadmin 192.168.2.1 Password: R2></pre> On PC A <pre>C:\>ssh -l SSHadmin 192.168.2.1 % Connection timed out; remote host not responding C:\></pre>
PART 3	Create a Numbered IPACL 120 on R1
Step 1	Verify that PC-C can access the PC-A via HTTPS using the web browser.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	
Step 2	Configure ACL 120 to specifically permit and deny the specified traffic. On R1 <pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 120 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 120 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 120 permit udp ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 120 permit udp any ? A.B.C.D Destination address any Any destination host eq Match only packets on a given port number gt Match only packets with a greater port number host A single destination host lt Match only packets with a lower port number neq Match only packets not on a given port number range Match only packets in the range of port numbers Router(config)#access-list 120 permit udp any host 192.136.1.3 ? dscp Match packets with given dscp value eq Match only packets on a given port number gt Match only packets with a greater port number lt Match only packets with a lower port number neq Match only packets not on a given port number precedence Match packets with given precedence value range Match only packets in the range of port numbers <cr> Router(config)#access-list 120 permit udp any host 192.136.1.3 eq ? <0-65535> Port number bootpc Bootstrap Protocol (BOOTP) client (68) bootps Bootstrap Protocol (BOOTP) server (67) domain Domain Name Service (DNS, 53) isakmp Internet Security Association and Key Management Protocol (500) non500-isakmp Internet Security Association and Key Management Protocol (4500) snmp Simple Network Management Protocol (161) tftp Trivial File Transfer Protocol (69) Router(config)#access-list 120 permit udp any host 192.136.1.3 eq domain</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 120 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 120 permit ?
ahp          Authentication Header Protocol
eigrp        Cisco's EIGRP routing protocol
esp          Encapsulation Security Payload
gre          Cisco's GRE tunneling
icmp         Internet Control Message Protocol
ip           Any Internet Protocol
ospf         OSPF routing protocol
tcp          Transmission Control Protocol
udp          User Datagram Protocol
Router(config)#access-list 120 permit tcp ?
A.B.C.D      Source address
any          Any source host
host         A single source host
Router(config)#access-list 120 permit tcp any ?
A.B.C.D      Destination address
any          Any destination host
eq           Match only packets on a given port number
gt           Match only packets with a greater port number
host         A single destination host
lt           Match only packets with a lower port number
neq          Match only packets not on a given port number
range        Match only packets in the range of port numbers
Router(config)#access-list 120 permit tcp any host 192.168.1.3 ?
dscp         Match packets with given dscp value
eq           Match only packets on a given port number
established   established
gt           Match only packets with a greater port number
lt           Match only packets with a lower port number
neq          Match only packets not on a given port number
precedence   Match packets with given precedence value
range        Match only packets in the range of port numbers
<cr>
Router(config)#access-list 120 permit tcp any host 192.168.1.3 eq ?
<0-65535>    Port number
ftp          File Transfer Protocol (21)
pop3         Post Office Protocol v3 (110)
smtp         Simple Mail Transport Protocol (25)
telnet       Telnet (23)
www          World Wide Web (HTTP, 80)
Router(config)#access-list 120 permit tcp any host 192.168.1.3 eq smtp
Router(config)#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 120 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 120 permit ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 120 permit tcp ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 120 permit tcp any ?
A.B.C.D     Destination address
any         Any destination host
eq          Match only packets on a given port number
gt          Match only packets with a greater port number
host        A single destination host
lt          Match only packets with a lower port number
neq         Match only packets not on a given port number
range       Match only packets in the range of port numbers
Router(config)#access-list 120 permit tcp any host 192.168.1.3 ?
dscp        Match packets with given dscp value
eq          Match only packets on a given port number
established established
gt          Match only packets with a greater port number
lt          Match only packets with a lower port number
neq         Match only packets not on a given port number
precedence  Match packets with given precedence value
range       Match only packets in the range of port numbers
<cr>
Router(config)#access-list 120 permit tcp any host 192.168.1.3 eq ?
<0-65535>   Port number
ftp         File Transfer Protocol (21)
pop3        Post Office Protocol v3 (110)
smtp        Simple Mail Transport Protocol (25)
telnet      Telnet (23)
www         World Wide Web (HTTP, 80)
Router(config)#access-list 120 permit tcp any host 192.168.1.3 eq ftp
Router(config)#
```

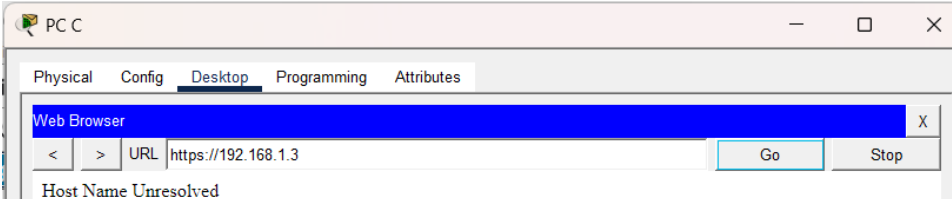
V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list 120 deny tcp ?
A.B.C.D Source address
any Any source host
host A single source host
Router(config)#access-list 120 deny tcp any ?
A.B.C.D Destination address
any Any destination host
eq Match only packets on a given port number
gt Match only packets with a greater port number
host A single destination host
lt Match only packets with a lower port number
neq Match only packets not on a given port number
range Match only packets in the range of port numbers
Router(config)#access-list 120 deny tcp any host ?
A.B.C.D Destination address
Router(config)#access-list 120 deny tcp any host 192.168.1.3 ?
dscp Match packets with given dscp value
eq Match only packets on a given port number
established established
gt Match only packets with a greater port number
lt Match only packets with a lower port number
neq Match only packets not on a given port number
precedence Match packets with given precedence value
range Match only packets in the range of port numbers
<cr>
Router(config)#access-list 120 deny tcp any host 192.168.1.3 eq ?
<0-65535> Port number
ftp File Transfer Protocol (21)
pop3 Post Office Protocol v3 (110)
smtp Simple Mail Transport Protocol (25)
telnet Telnet (23)
www World Wide Web (HTTP, 80)
Router(config)#access-list 120 deny tcp any host 192.168.1.3 eq 443
Router(config)#
```

Name: Dhruv N Chandalia

	<pre> Router(config)#access-list 120 deny tcp any host 192.168.3.3 eq 22 <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 120 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 120 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 120 permit tcp ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 120 permit tcp host ? A.B.C.D Source address Router(config)#access-list 120 permit tcp host 192.168.3.3 ? A.B.C.D Destination address any Any destination host eq Match only packets on a given port number gt Match only packets with a greater port number host A single destination host lt Match only packets with a lower port number neq Match only packets not on a given port number range Match only packets in the range of port numbers Router(config)#access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 ? dscp Match packets with given dscp value eq Match only packets on a given port number established established gt Match only packets with a greater port number lt Match only packets with a lower port number neq Match only packets not on a given port number precedence Match packets with given precedence value range Match only packets in the range of port numbers <cr> Router(config)#access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq ? <0-65535> Port number ftp File Transfer Protocol (21) pop3 Post Office Protocol v3 (110) smtp Simple Mail Transport Protocol (25) telnet Telnet (23) www World Wide Web (HTTP, 80) Router(config)#access-list 120 permit tcp host 192.168.3.3 host 10.1.1.1 eq 22 Router(config)# </pre>
Coop	
Step 3	Apply ACL to interface S0/0/0 <pre> Router(config)#interface s0/0/0 Router(config-if)#ip access-group 120 in Router(config-if)# </pre>
Step 4	Verify that PC-C cannot access PC-A via HTTPS using the web browser.  The screenshot shows a window titled 'PC C' with a 'Web Browser' tab. The address bar contains the URL 'https://192.168.1.3'. Below the address bar, the text 'Host Name Unresolved' is displayed, indicating that the browser cannot reach the destination.

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

PART 5	Create a Numbered IP ACL 110 on R3
Step 1	<pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 110 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 110 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 110 permit ip ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 110 permit ip 192.168.3.0 ? A.B.C.D Source wildcard bits Router(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 ? A.B.C.D Destination address any Any destination host host A single destination host Router(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 any Router(config)#</pre>
Step 2	Apply <pre>Router(config)#interface g0/1 Router(config-if)#ip access-group 110 in Router(config-if)#</pre>
Part 6	Create a Numbered IP ACL 100 on R3
Step 1	On R3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 permit ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 100 permit tcp ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 100 permit tcp 10.0.0.0
% Incomplete command.
Router(config)#access-list 100 permit tcp 10.0.0.0?
A.B.C.D
Router(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 ?
A.B.C.D     Destination address
any         Any destination host
eq          Match only packets on a given port number
gt          Match only packets with a greater port number
host        A single destination host
lt          Match only packets with a lower port number
neq         Match only packets not on a given port number
range       Match only packets in the range of port numbers
Router(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq ?
<0-65535>   Port number
ftp         File Transfer Protocol (21)
pop3        Post Office Protocol v3 (110)
smtp        Simple Mail Transport Protocol (25)
telnet      Telnet (23)
www         World Wide Web (HTTP, 80)
Router(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host ?
A.B.C.D     Destination address
Router(config)#access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 deny ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 100 deny ip ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 100 deny ip 10.0.0.0 ?
A.B.C.D     Source wildcard bits
Router(config)#access-list 100 deny ip 10.0.0.0 0.255.255.255 ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 100 deny ip 10.0.0.0 0.255.255.255 any
Router(config)#
```

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 deny ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 100 deny ip ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 100 deny ip 172.16.0.0 ?
A.B.C.D     Source wildcard bits
Router(config)#access-list 100 deny ip 172.16.0.0 0.15.255.255 ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 100 deny ip 172.16.0.0 0.15.255.255 any
Router(config)#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 deny ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 100 deny ip ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 100 deny ip 192.168.0.0 ?
A.B.C.D     Source wildcard bits
Router(config)#access-list 100 deny ip 192.168.0.0 0.0.255.255 ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 100 deny ip 192.168.0.0 0.0.255.255 any
Router(config)#
```

```
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 deny ?
ahp         Authentication Header Protocol
eigrp       Cisco's EIGRP routing protocol
esp         Encapsulation Security Payload
gre         Cisco's GRE tunneling
icmp        Internet Control Message Protocol
ip          Any Internet Protocol
ospf        OSPF routing protocol
tcp         Transmission Control Protocol
udp         User Datagram Protocol
Router(config)#access-list 100 deny ip ?
A.B.C.D     Source address
any         Any source host
host        A single source host
Router(config)#access-list 100 deny ip 127.0.0.0 ?
A.B.C.D     Source wildcard bits
Router(config)#access-list 100 deny ip 127.0.0.0 0.255.255.255 ?
A.B.C.D     Destination address
any         Any destination host
host        A single destination host
Router(config)#access-list 100 deny ip 127.0.0.0 0.255.255.255 any
Router(config)#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router(config)#access-list ?
% Unrecognized command
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 100 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 100 deny ?
ahp          Authentication Header Protocol
eigrp        Cisco's EIGRP routing protocol
esp          Encapsulation Security Payload
gre          Cisco's GRE tunneling
icmp         Internet Control Message Protocol
ip           Any Internet Protocol
ospf         OSPF routing protocol
tcp          Transmission Control Protocol
udp          User Datagram Protocol
Router(config)#access-list 100 deny 224.0.0.0 ?
% Unrecognized command
Router(config)#access-list 100 deny ip ?
A.B.C.D      Source address
any          Any source host
host         A single source host
Router(config)#access-list 100 deny ip 224.0.0.0 ?
A.B.C.D      Source wildcard bits
Router(config)#access-list 100 deny ip 224.0.0.0 15.255.255.255 ?
A.B.C.D      Destination address
any          Any destination host
host         A single destination host
Router(config)#access-list 100 deny ip 224.0.0.0 15.255.255.255 any
Router(config)#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 100 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 100 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 100 permit ip ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 100 permit ip any ? A.B.C.D Destination address any Any destination host host A single destination host Router(config)#access-list 100 permit ip any any Router(config)#</pre>
Step 2	Apply the ACL to interface Serial 0/0/1. Use the ip access-group command to apply the access list to incoming traffic on interface Serial 0/0/1 <pre>Router(config)#interface s0/0/1 Router(config-if)#ip access-group 100 in Router(config-if)#</pre>
Step 3	Confirm that the specified traffic entering interface Serial 0/0/1 is handled correctly. <pre>conf t no access-list 100 ! access-list 100 permit tcp host 192.168.2.1 eq 22 host 192.168.3.3 access-list 100 permit tcp 10.0.0.0 0.255.255.255 eq 22 host 192.168.3.3 access-list 100 deny ip 10.0.0.0 0.255.255.255 any access-list 100 deny ip 172.16.0.0 0.15.255.255 any access-list 100 deny ip 192.168.0.0 0.0.255.255 any access-list 100 deny ip 127.0.0.0 0.255.255.255 any access-list 100 deny ip 224.0.0.0 15.255.255.255 any access-list 100 permit ip any any end</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

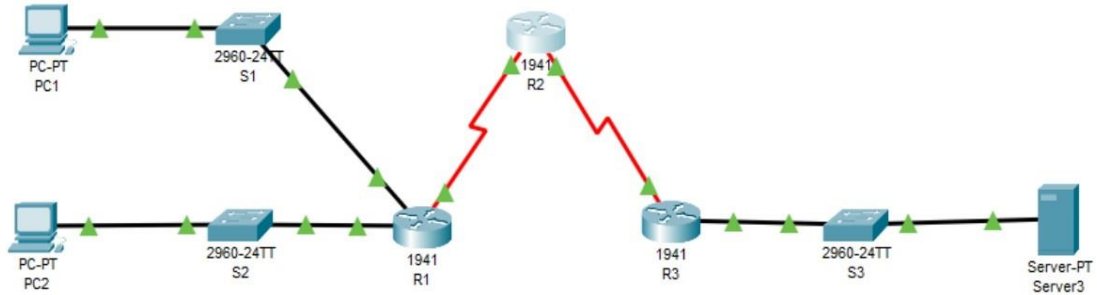
	Wt
Step 4	Verify On PC C C:\>ping 192.168.1.3 Pinging 192.168.1.3 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 192.168.1.3: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), C:\>ssh -l SSHadmin 192.168.2.1 Password: R2>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

5. Configuring IPv6 ACLs



Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
Server3	NIC	2001:DB8:1:30::30/64	FE80::30

```
R1(config)#
R1(config)#enable secret enpa55
R1(config)#
R1(config)#int gig0/0
R1(config-if)#ipv6 address 2001:DB8:1:10::1/64
R1(config-if)#ipv6 address FE80::1 link-local
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#int gig0/1
R1(config-if)#ipv6 address 2001:DB8:1:11::1/64
R1(config-if)#ipv6 address FE80::1 link-local
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
EXIT
R1(config)#int se0/1/0
R1(config-if)#ipv6 address 2001:DB8:1:1::1/64
R1(config-if)#ipv6 address FE80::1 link-local
R1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#exit
R1(config)#

R2(config)#int se0/1/0
R2(config-if)#ipv6 address 2001:DB8:1:1::2/64
R2(config-if)#ipv6 address FE80::2 link-local
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
exit
R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

R2(config)#int se0/1/1
R2(config-if)#ipv6 address 2001:DB8:1:2::2/64
R2(config-if)#ipv6 address FE80::2 link-local
R2(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
R2(config-if)#exit
R2(config)#
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>R3(config)#enable secret enpa55 R3(config)#int gig0/0 R3(config-if)#ipv6 address 2001:DB8:1:30::1/64 R3(config-if)#ipv6 address FE80::3 link-local R3(config-if)#no shut R3(config-if)# %LINK-S-CHANGED: Interface GigabitEthernet0/0, changed state to up %LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up exit R3(config)#int se0/1/0 R3(config-if)#ipv6 address 2001:DB8:1:2::1/64 R3(config-if)#ipv6 address FE80::3 link-local R3(config-if)#no shut R3(config-if)# %LINK-S-CHANGED: Interface Serial0/1/0, changed state to up R3(config-if)#exit %LINEPROTO-S-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up</pre> Step 2: Routing <pre>R1(config)# R1(config)#ipv6 unicast-routing R1(config)#ipv6 route 2001:DB8:1:2::0/64 2001:DB8:1:1::2 R1(config)#ipv6 route 2001:DB8:1:30::0/64 2001:DB8:1:1::2 R1(config)# R2(config)# R2(config)#ipv6 unicast-routing R2(config)#ipv6 route 2001:DB8:1:10::0/64 2001:DB8:1:1::1 R2(config)#ipv6 route 2001:DB8:1:11::0/64 2001:DB8:1:1::1 R2(config)#ipv6 route 2001:DB8:1:30::0/64 2001:DB8:1:2::1 R2(config)# R3(config)#ipv6 unicast-routing R3(config)#ipv6 route 2001:DB8:1:10::0/64 2001:DB8:1:2::2 R3(config)#ipv6 route 2001:DB8:1:11::0/64 2001:DB8:1:2::2 R3(config)#ipv6 route 2001:DB8:1:1::0/64 2001:DB8:1:2::2</pre> Step 3: Verify connection <pre>PC1>ping 2001:DB8:1:30::30 PC2>ping 2001:DB8:1:30::30</pre>
	Part 1: Configure, apply and verify an IPv6 ACL. Step 1: Configure an ACL that will block HTTP and HTTPS access. <pre>R1(config)#ipv6 access-list BLOCK_HTTP R1(config-ipv6-acl)# deny tcp any host 2001:DB8:1:30::30 eq www R1(config-ipv6-acl)# deny tcp any host 2001:DB8:1:30::30 eq 443 R1(config-ipv6-acl)# permit ipv6 any any R1(config-ipv6-acl)#exit</pre> Step 2 : Apply the ACL to the correct interface. <pre>R1(config)# int gig0/1 R1(config-if)# ipv6 traffic-filter BLOCK_HTTP in</pre> Step 3: Verify the ACL implementation. • Open the web browser of PC1 to display the webpage. Desktop>web browser> http://2001:DB8:1:30::30 Desktop>web browser> https://2001:DB8:1:30::30 • Open the web browser of PC2 to display the webpage. Desktop>web browser> http://2001:DB8:1:30::30 Desktop>web browser> https://2001:DB8:1:30::30 • PC2>ping 2001:DB8:1:30::30
	Part 2: Configure, Apply, and Verify a Second IPv6 ACL Step 1: Create an access list to block ICMP. <pre>R3(config)#ipv6 access-list BLOCK_ICMP R3(config-ipv6-acl)# deny icmp any any R3(config-ipv6-acl)# permit ipv6 any any</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>R3(config-ipv6-acl)#exit Step 2 : Apply the ACL to the correct interface. R1(config)# int gig0/0 R1(config-if)# ipv6 traffic-filter BLOCK_ICMP out Step 3:Verify that the proper access-list functions.</pre> • Ping from PC2 to 2001:DB8:1:30::30. • Ping from PC1 to 2001:DB8:1:30::30. • Open the web browser of PC1 to http://2001:DB8:1:30::30 or https://2001:DB8:1:30::30.
--	---

V. G. VAZE KELKAR COLLEGE, MULUND (E)

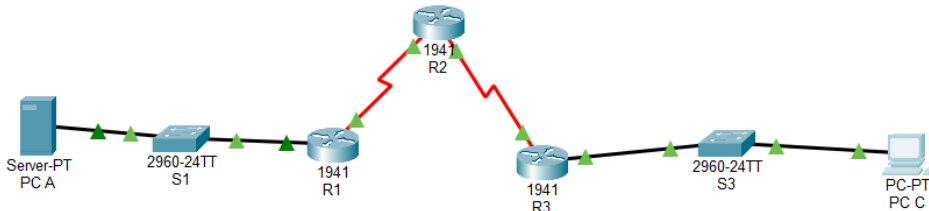
Roll No: A004

Name: Dhruv N Chandalia

Pracytical 5

Configuring a Zone-Based Policy Firewall (ZPF)

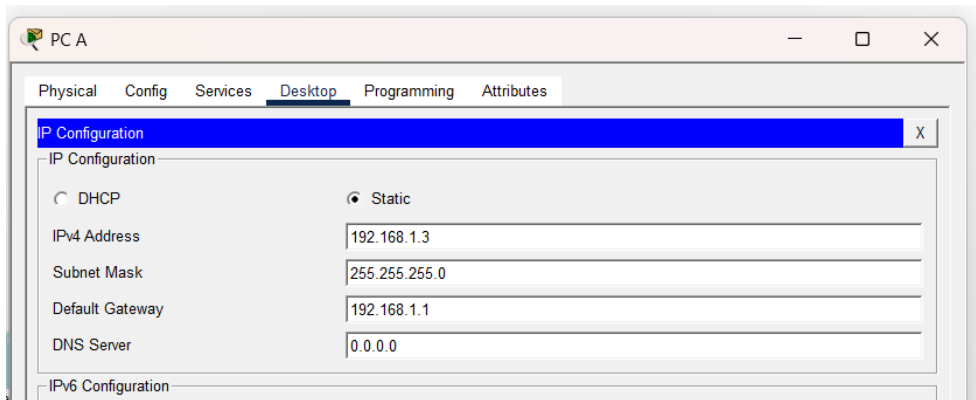
TOPOLOGY



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

PC A

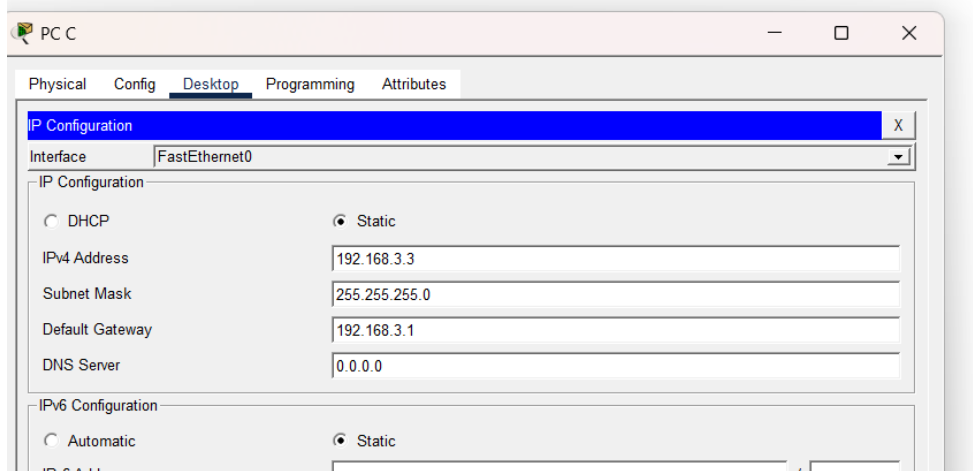


PC C

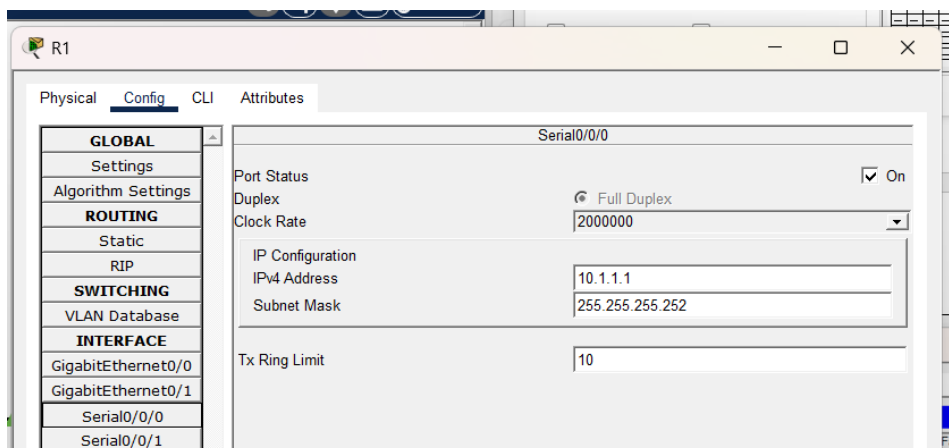
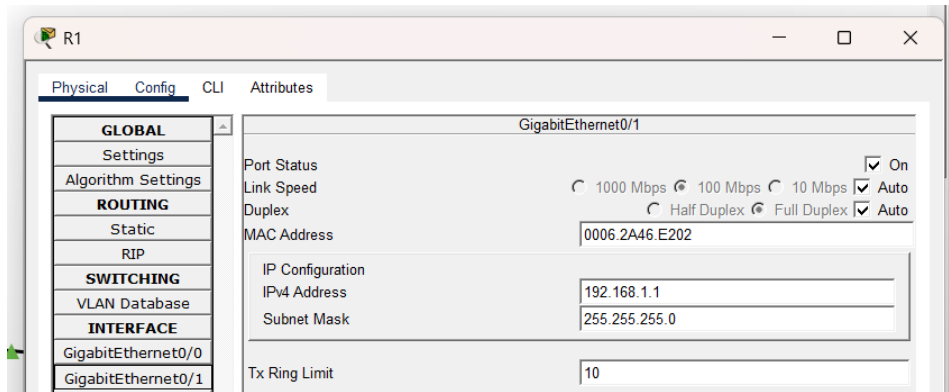
V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



R1



R2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.1.1.2

Subnet Mask 255.255.255.252

Tx Ring Limit 10

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/0/1

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.2.2.2

Subnet Mask 255.255.255.252

Tx Ring Limit 10

R3

R3

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0009.7C7D.5202

IP Configuration

IPv4 Address 192.168.3.1

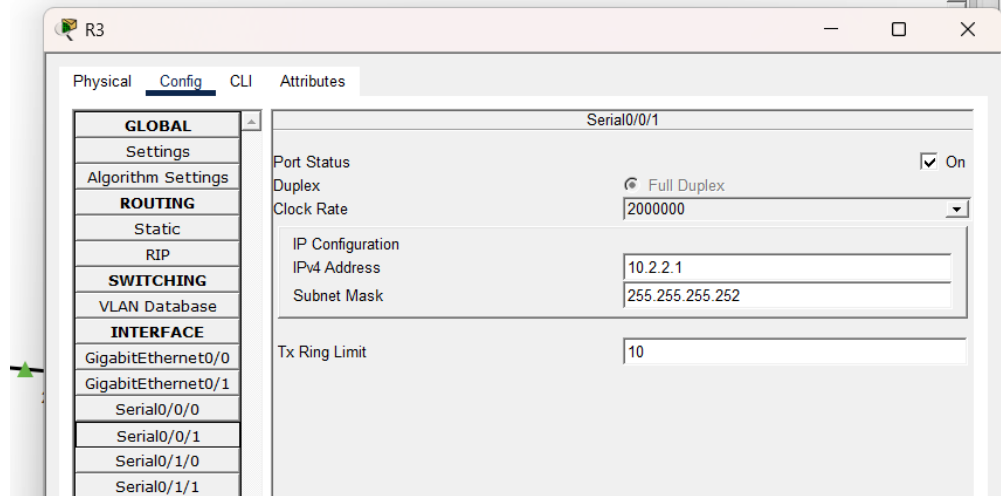
Subnet Mask 255.255.255.0

Tx Ring Limit 10

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	
	OSPF R1 <pre>Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router ospf 1 Router(config-router)#network 192.168.1.0 0.0.0.255 area 0 Router(config-router)#network 10.1.1.0 0.0.0.3 area 0 Router(config-router)#end Router# %SYS-5-CONFIG_I: Configured from console by console 00:30:08: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL, Loading Done</pre> R2 <pre>%SYS-5-CONFIG_I: Configured from console by console conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router ospf 1 Router(config-router)#network 10.1.1.0 0.0.0.3 area 0 Router(config-router)#network 10.2.2.0 0.0.0.3 area 0 Router(config-router)#end Router# %SYS-5-CONFIG_I: Configured from console by console</pre> R3 <pre>Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router ospf 1 Router(config-router)#network 192.168.3.0 0.0.0.255 area 0 Router(config-router)#network 10.2.2.0 0.0.0.3 area 0 Router(config-router)#end Router# %SYS-5-CONFIG_I: Configured from console by console</pre>
Part 1	Verify Basic Network Connectivity

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

Step 1: From the PC-A command prompt, ping PC-C at 192.168.3.3.

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

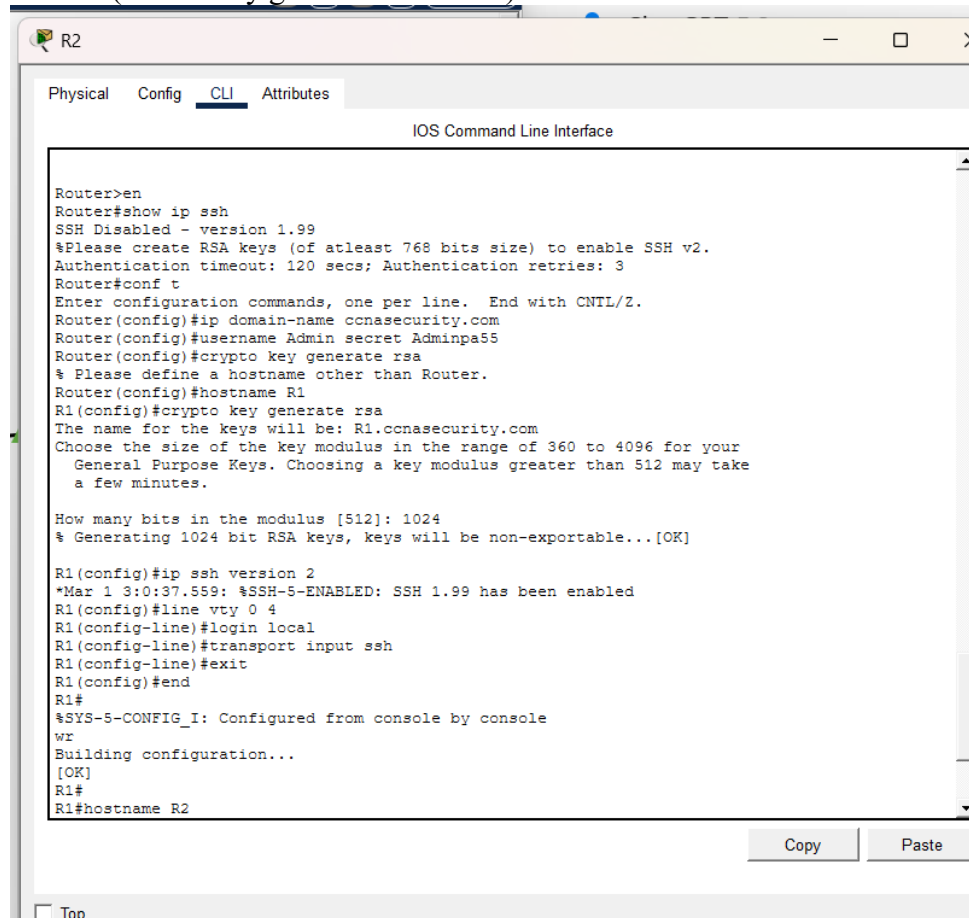
Reply from 192.168.3.3: bytes=32 time=1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=5ms TTL=128
Reply from 192.168.3.3: bytes=32 time=3ms TTL=128
Reply from 192.168.3.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 3ms

C:\>
```

Step 2

Step 2: Access R2 using SSH.
On R2 (mistakenly gave hostname R1)



```
R2
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Router#show ip ssh
SSH Disabled - version 1.99
%Please create RSA keys (of atleast 768 bits size) to enable SSH v2.
Authentication timeout: 120 secs; Authentication retries: 3
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-name ccnasecurity.com
Router(config)#username Admin secret Adminpa55
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#hostname R1
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

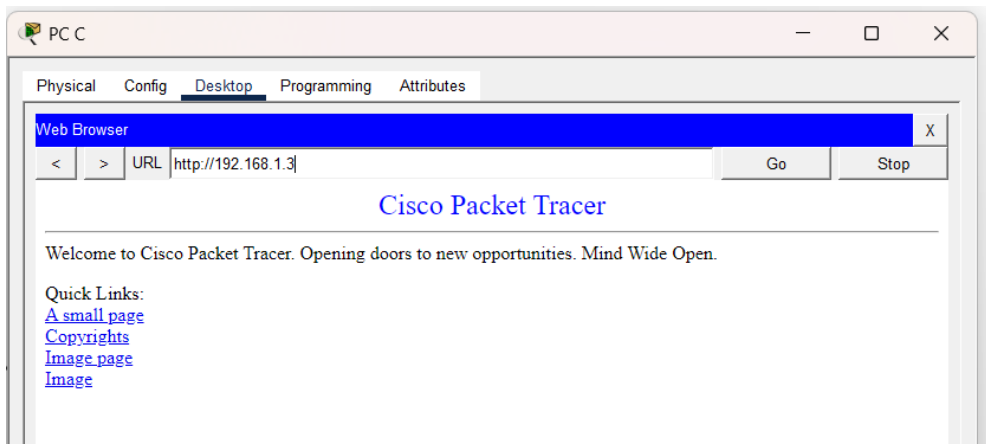
R1(config)#ip ssh version 2
*Mar 1 3:0:37.559: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
R1#
R1#hostname R2
```

At PC C

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>C:\>ssh -l Admin 10.2.2.2 Password: R1></pre>
Step 3	Step 3: From PC-C, open a web browser to the PC-A server. 
PART 2	Create the Firewall Zones on R3
Step 1	Enable Security Technology Package <pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#license boot module c1900 technology-package securityk9 PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE OR LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING SUCH PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE FOLLOWING TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE BOUND BY ALL THE TERMS SET FORTH HEREIN. Use of this product feature requires an additional license from Cisco, together with an additional payment. You may use this product feature on an evaluation basis, without payment to Cisco, for 60 days. Your use of the product, including during the 60 day evaluation period, is subject to the Cisco end user license agreement http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html If you use the product feature beyond the 60 day evaluation period, you must submit the appropriate payment to Cisco for the license. After the 60 day evaluation period, your use of the product feature will be governed solely by the Cisco end user license agreement (link above), which can be found at the following URL: http://www.cisco.com/en/US/docs/general/warranty/English/EU1KEN_.html Reload command to save the package</pre>

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>Router#reload Proceed with reload? [confirm] System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1) Technical Support: http://www.cisco.com/techsupport Copyright (c) 2010 by cisco Systems, Inc. Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB CISCO1941/K9 platform with 524288 Kbytes of main memory Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled Readonly ROMMON initialized program load complete, entry point: 0x80803000, size: 0x1b340 program load complete, entry point: 0x80803000, size: 0x1b340 IOS Image Load Test Digitally Signed Release Software program load complete, entry point: 0x81000000, size: 0x2bb1c58 Self decompressing the image : ##### [OK] Smart Init is enabled smart init is sizing iomem TYPE MEMORY_REQ HWIC Slot 0 0x00200000 HWIC Slot 1 0x00200000 Onboard devices & buffer pools 0x01E8F000 ----- TOTAL: 0x02E8F000 Rounded IOMEM up to: 48Mb. Using 6 percent iomem. [48Mb/512Mb]</pre>
Step 2 3	<pre>Router# Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#zone security IN-ZONE Router(config-sec-zone)#exit Router(config)#zone security OUT-ZONE Router(config-sec-zone)#exit Router(config)#</pre>
PART 3	At R3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	<pre>Router>en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#access-list ? <1-99> IP standard access list <100-199> IP extended access list Router(config)#access-list 101 ? deny Specify packets to reject permit Specify packets to forward remark Access list entry comment Router(config)#access-list 101 permit ? ahp Authentication Header Protocol eigrp Cisco's EIGRP routing protocol esp Encapsulation Security Payload gre Cisco's GRE tunneling icmp Internet Control Message Protocol ip Any Internet Protocol ospf OSPF routing protocol tcp Transmission Control Protocol udp User Datagram Protocol Router(config)#access-list 101 permit ip ? A.B.C.D Source address any Any source host host A single source host Router(config)#access-list 101 permit ip 192.168.3.0 ? A.B.C.D Source wildcard bits Router(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 ? A.B.C.D Destination address any Any destination host host A single destination host Router(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 any Router(config)#</pre>
	Permitting all traffic from PC C
PART 4	<pre>Router(config)#policy-map type inspect IN-OUT-MAP Router(config-pmap)#class type inspect IN-NET-CLASS-MAP Router(config-pmap-c)#inspect %No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All protocols will be inspected Router(config-pmap-c)#exit Router(config-pmap)#exit Router(config)#</pre>
PART 5	<pre>Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#policy-map type inspect IN-2-OUT-PMAP Router(config-pmap)# class type inspect IN-NET-CLASS-MAP Router(config-pmap-c)# inspect %No specific protocol configured in class IN-NET-CLASS-MAP for inspection. All protocols will be inspected Router(config-pmap-c)# exit Router(config-pmap)#exit Router(config)#zone-pair security IN-2-OUT-2PAIR source IN-ZONE destination OUT-ZONE Router(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PMAP Router(config-sec-zone-pair)#exit Router(config)#interface g0/1 Router(config-if)#zone-member security IN-ZONE Router(config-if)#exit Router(config)# Router(config)#interface s0/0/1 Router(config-if)#zone-member security OUT-ZONE Router(config-if)#exit Router(config)#</pre>
PART 6	

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=8ms TTL=125
Reply from 192.168.1.3: bytes=32 time=7ms TTL=125
Reply from 192.168.1.3: bytes=32 time=8ms TTL=125
Reply from 192.168.1.3: bytes=32 time=9ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 9ms, Average = 8ms

C:\>Part 6: Test Firewall Functionality from IN-ZONE to OUT-ZONE. Verify that internal hosts
can still access external resources after configuring the ZPF. Step 1: From internal PC-C,
ping the external PC-A server.
Invalid Command.

C:\>ssh -l Admin 10.2.2.2

Password:
% Login invalid

Password:
```

```
Router#show policy-map type inspect zone-pair sessions

policy exists on zp IN-2-OUT-ZPAIR
Zone-pair: IN-2-OUT-ZPAIR

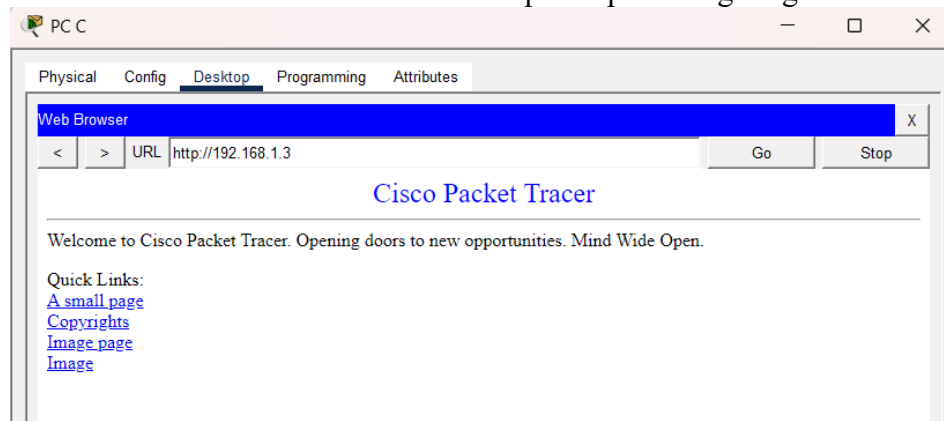
Service-policy inspect : IN-2-OUT-PMAP

Class-map: IN-NET-CLASS-MAP (match-all)
  Match: access-group 101
  Inspect

    Number of Established Sessions = 1
    Established Sessions
      Session 73139328 (192.168.3.3:1027)=>(10.2.2.2:22) tcp SIS_OPEN/TCP_ESTAB
        Created 00:00:13, Last heard 00:00:08
        Bytes sent (initiator:responder) [1064:895]
    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes

Router#
```

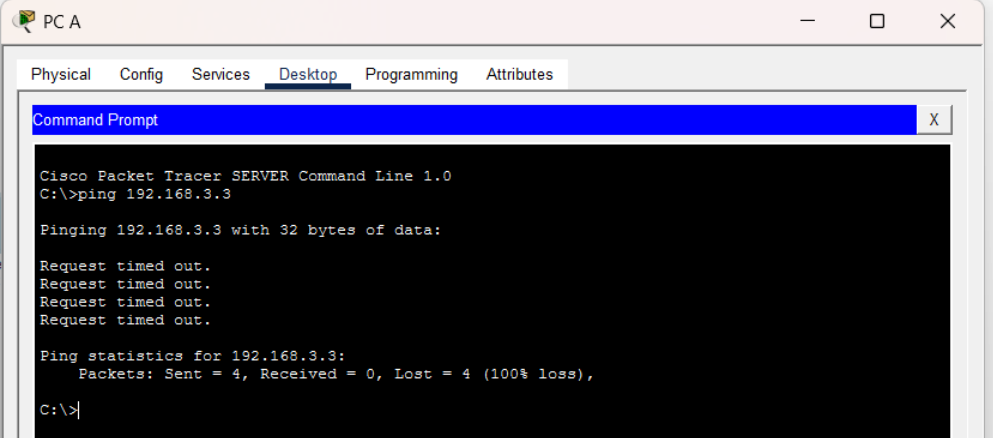
End ssh session in order to check the http. Keep clicking on go to create sessions.



V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

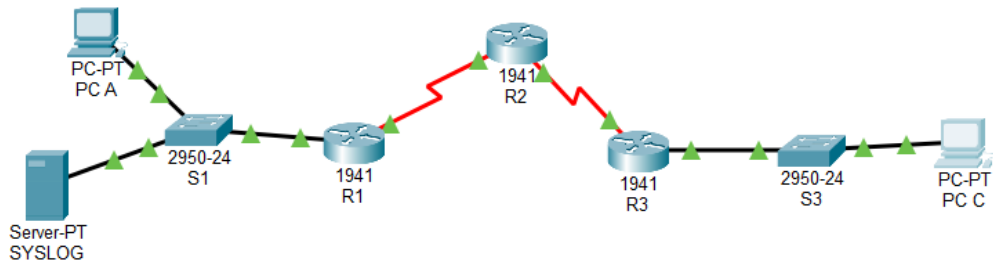
	<pre>Router#show policy-map type inspect zone-pair sessions policy exists on zp IN-2-OUT-ZPAIR Zone-pair: IN-2-OUT-ZPAIR Service-policy inspect : IN-2-OUT-PMAP Class-map: IN-NET-CLASS-MAP (match-all) Match: access-group 101 Inspect Number of Established Sessions = 10 Established Sessions Session 200567408 (192.168.3.3:1031)=>(192.168.1.3:80) tcp SIS_OPEN/TCP_ESTAB Created 00:00:03, Last heard 00:00:03 Bytes sent (initiator:responder) [284:575] Session 200567920 (192.168.3.3:1032)=>(192.168.1.3:80) tcp SIS_OPEN/TCP_ESTAB Created 00:00:02, Last heard 00:00:02 Bytes sent (initiator:responder) [284:575] Session 200569456 (192.168.3.3:1033)=>(192.168.1.3:80) tcp SIS_OPEN/TCP_ESTAB Created 00:00:02, Last heard 00:00:02 Bytes sent (initiator:responder) [284:575] Session 200568432 (192.168.3.3:1034)=>(192.168.1.3:80) tcp SIS_OPEN/TCP_ESTAB --More--</pre>
PART 7	 <pre>PC A Physical Config Services Desktop Programming Attributes Command Prompt Cisco Packet Tracer SERVER Command Line 1.0 C:\>ping 192.168.3.3 Pinging 192.168.3.3 with 32 bytes of data: Request timed out. Request timed out. Request timed out. Request timed out. Ping statistics for 192.168.3.3: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), C:\></pre> <pre>R2#ping 192.168.3.3 Type escape sequence to abort. Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds: Success rate is 0 percent (0/5) R2#</pre>

Practical 6	Configure IOS Intrusion Prevention System(IPS)Using the CLI
	TOPOLOGY

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

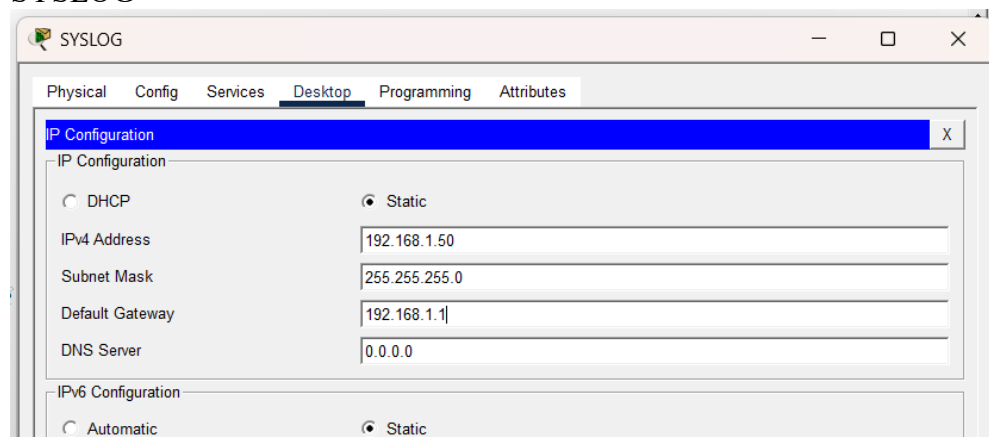
Name: Dhruv N Chandalia



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/1
	S0/0/0	10.2.2.1	255.255.255.252	N/A	N/A
Syslog	NIC	192.168.1.50	255.255.255.0	192.168.1.1	S1 F0/2
PC-A	NIC	192.168.1.2	255.255.255.0	192.168.1.1	S1 F0/3
PC-C	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/2

SYSLOG



PC A

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

PC A

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

PC C

PC C

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.3.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.3.1

DNS Server 0.0.0.0

R1

R1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.1.1.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

R1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

GigabitEthernet0/1

Port Status ☒ On

Link Speed ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0050.0F85.E402

IP Configuration

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

R2

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

Serial0/0/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.1.1.2

Subnet Mask 255.255.255.252

Tx Ring Limit 10

R2

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/1

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 2000000

IP Configuration

IPv4 Address 10.2.2.2

Subnet Mask 255.255.255.252

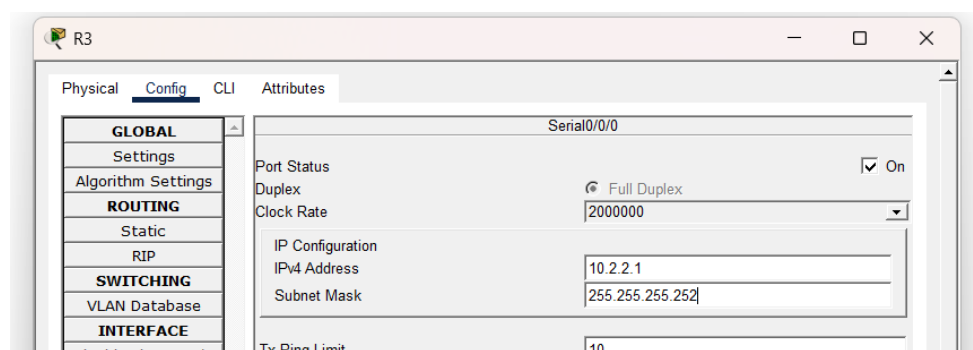
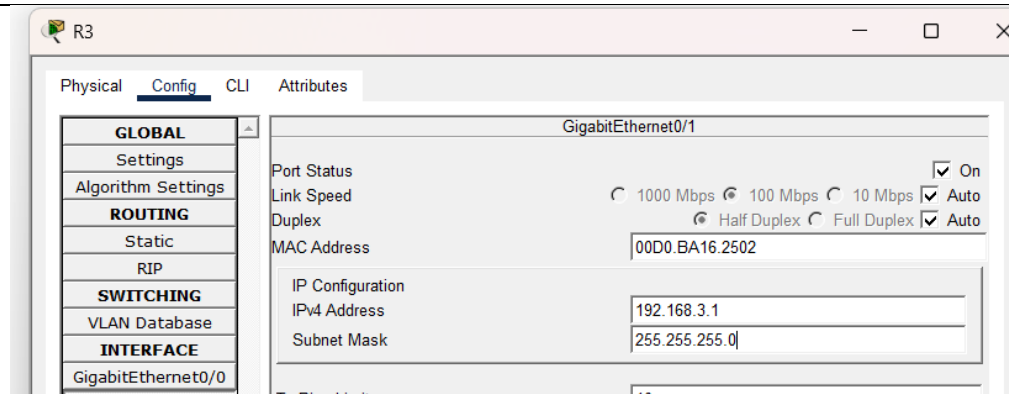
Tx Ring Limit 10

R3

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia



PART 1

Enable IOS IPS

```
Router(config)#license boot module c1900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING SUCH
PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.
```

Step 1

On R1

Use write command to save.

#show version

```
Technology Package License Information for Module:'c1900'
```

Technology	Technology-package Current	Technology-package Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	None	None	securityk9
--More--			

Step 2:

Perform ospf

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.255 area 0
Router(config-router)#network 10.1.1.0 0.0.0.3 area 0
Router(config-router)#exit
Router(config)#
00:14:55: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0
```

R2

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#router ospf 1
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#network
00:14:47: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL,
Loading Done

% Incomplete command.
R2(config-router)#network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#
00:17:04: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from LOADING to FULL,
Loading Done
```

R3

```
Router(config)#hostname R3
R3(config)#router ospf 1
R3(config-router)#network 192.168.3.0 0.0.0.3 area 0
R3(config-router)#no network 192.168.3.0 0.0.0.3 area 0
R3(config-router)#network 10.2.2.0 0.0.0.3 area 0
R3(config-router)#
00:16:56: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL,
Loading Done

R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#
*Mar16/2020 5:49:00PM: Line protocol on Interface Serial0/0/0, changed state to down
```

Verify

PC A to PC C

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=10ms TTL=125
Reply from 192.168.3.2: bytes=32 time=8ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 5ms
```

PC C to PC A

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>
```

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

	Step 3 <pre>R1#mkdir ipsdir Create directory filename [ipsdir]? Created dir flash:ipsdir</pre> Step 4 <pre>R1#show version Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2) Technical Support: http://www.cisco.com/techsupport</pre> #reload Then <pre>R1#conf t Enter configuration commands, one per line. End with CNTL/Z. R1(config)#ip ips config location flash:ipsdir R1(config)#</pre> <pre>R1#conf t Enter configuration commands, one per line. End with CNTL/Z. R1(config)#ip ips config location flash:ipsdir R1(config)#ip ips name iosips R1(config)#ip ips notify log R1(config)#exit R1# %SYS-5-CONFIG_I: Configured from console by console</pre> <pre>R1#clock set 18:40:00 20 February 2026 R1#conf t Enter configuration commands, one per line. End with CNTL/Z. R1(config)#service timestamps log datetime msec R1(config)#logging host 192.168.1.50 R1(config)#</pre>
	Step 7 <pre>R1#clock set 18:40:00 20 February 2026 R1#conf t Enter configuration commands, one per line. End with CNTL/Z. R1(config)#service timestamps log datetime msec R1(config)#logging host 192.168.1.50 R1(config)#ip ips signature-category R1(config-ips-category)#category all R1(config-ips-category-action)#retired true R1(config-ips-category-action)#exit R1(config-ips-category)#category ios_ips basic R1(config-ips-category-action)#retired false R1(config-ips-category-action)#exit R1(config-ips-category)#exit Do you want to accept these changes? [confirm] Applying Category configuration to signatures ... %IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines %IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be scanned</pre> R1(config)# Step 8 <pre>R1(config)#interface g0/1 R1(config-if)#ip ips iosips out R1(config-if)#</pre> <pre>*Feb 20, 18:58:02.5858: %IPS-6-ENGINE_BUILDS_STARTED: 18:58:02 UTC Feb 20 2026 *Feb 20, 18:58:02.5858: %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13 engines *Feb 20, 18:58:02.5858: %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for this engine will be scanned *Feb 20, 18:58:02.5858: %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms</pre>
PART 2	Modify Signature

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

Step 1:

```
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#enabled true
R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action deny-packet-inline
R1(config-sigdef-sig-engine)#exit
R1(config-sigdef-sig)#exit
R1(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms

R1(config)#
```

Step 2:

```

R1#show ip ips all
IPS Signature File Configuration Status
  Configured Config Locations: flash:ipsdir
  Last signature default load time:
  Last signature delta load time:
  Last event action (SEAP) load time: -none-

  General SEAP Config:
  Global Deny Timeout: 3600 seconds
  Global Overrides Status: Enabled
  Global Filters Status: Enabled

IPS Auto Update is not currently configured

IPS Status and SDEF Notification Status
```

Step 3:

From PC C to PC A

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

From PC A to PC C

V. G. VAZE KELKAR COLLEGE, MULUND (E)

Roll No: A004

Name: Dhruv N Chandalia

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=12ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=8ms TTL=125
Reply from 192.168.3.2: bytes=32 time=8ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 12ms, Average = 7ms

C:\>
```

Step 4

The screenshot shows the 'Syslog' configuration window in a network device interface. The 'Services' tab is selected, and the 'Syslog' service is enabled. A table shows 11 log entries with columns for Time, HostName, and Message. The messages include OSPF and IPS status updates.

	Time	HostName	Message
1	02.20.2026 06:54:17.200 PM	192.168.1.1	%LINEPROTO-5-UPDOWN...
2	02.20.2026 06:54:27.207 PM	192.168.1.1	18:54:27: %OSPF-5-...
3	02.20.2026 06:58:02.981 PM	192.168.1.1	%IPS-6-...
4	02.20.2026 06:58:02.981 PM	192.168.1.1	%IPS-6-...
5	02.20.2026 06:58:02.981 PM	192.168.1.1	%IPS-6-ENGINE_READY...
6	02.20.2026 06:58:02.981 PM	192.168.1.1	%IPS-6-...
7	02.20.2026 07:12:13.165 PM	192.168.1.1	%LINEPROTO-5-UPDOWN...
8	02.20.2026 07:12:13.165 PM	192.168.1.1	19:12:13: %OSPF-5-...
9	02.20.2026 07:12:13.165 PM	192.168.1.1	%LINEPROTO-5-UPDOWN...
10	02.20.2026 07:12:23.176 PM	192.168.1.1	19:12:23: %OSPF-5-...
11	02.20.2026 07:21:38.242 PM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console b...